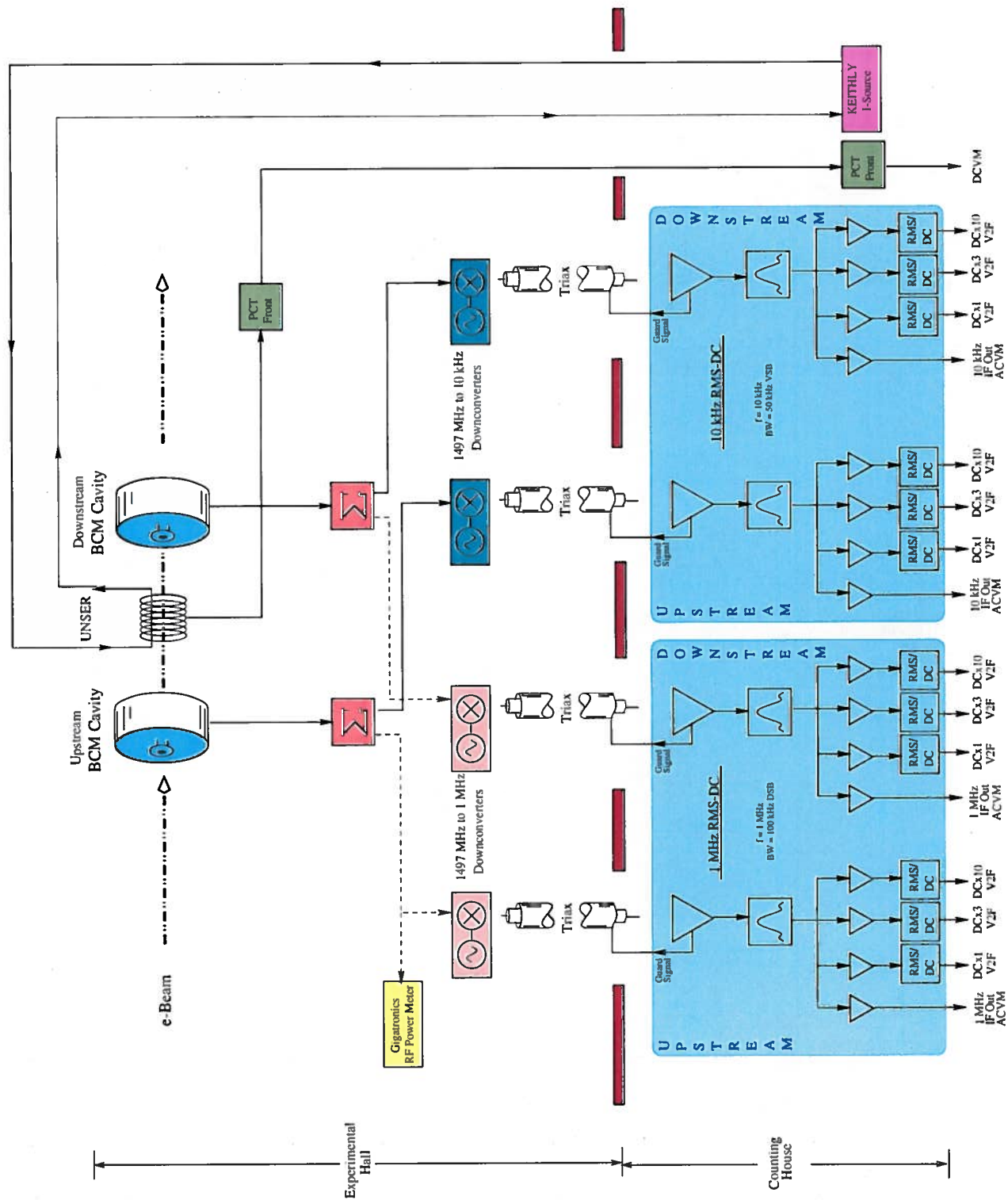


Hall A BCM System



EUROPEAN ORGANIZATION FOR NUCLEAR RESEARCH

CERN-ISR-OP/81-14

A TORROIDAL D.C. BEAM CURRENT TRANSFORMER WITH HIGH RESOLUTION

by

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Summary

Beam current is measured with a system combining a second harmonic magnetic modulator with an active current transformer in an operational feedback loop, to obtain wide band response down to dc. A beam current monitor using this principle has been in operation in the CERN Intersecting Storage Rings (ISR) for over 10 years. The original design concept and performance has since been improved. The magnetic modulator-demodulator circuit has two modes of operation: low gain and wide band for initial search until tracking (with the beam current signal) and high gain, narrow band for optimum dc stability during normal operation. A second active L/R integrator loop is added to remove any residual modulator noise from the output signal to improve signal-to-noise ratio in the mid-frequency range. The performance of the system depends very much on the magnetic characteristics of the toroidal cores. The choice of material, measurement methods and core selection is discussed. More recent applications are beam intensity monitors for the CERN Antiproton Accumulator Ring (AA) and the CERN Proton Synchrotron (CPS).

Historical background

Early in 1961, the circulating beam in the CERN Proton Synchrotron (PS) was measured with an active current transformer circuit¹ originally proposed by H. Hereward. The main feature of the "Hereward" transformer was a beam sensing toroid transformer with bifilar windings in the feedback loop of a high gain operational amplifier (L/R integrator scheme). This provided an extended low frequency response far into the sub-cycle range. An improved version² of this monitor, with separate high and low frequency toroid transformers was built in 1967.

These ac beam transformer systems provided a quasi dc response during the short (2...5 s) PS machine cycle. However, they needed periodic resetting to zero in synchronism with the PS cycle to erase accumulated integrator drift.

In order to eliminate the actual cause of this drift, the author considered the use of a 2nd harmonic magnetic modulator (a well-known magnetic amplifier technique) and it soon became apparent that the combination³ of an active current transformer and a magnetic modulator in a common feedback loop could approach the performance of an ideal dc current transformer in many applications.

Operating principle

The combined beam current transformer consists of a number of toroidal cores, strip wound from a high permeability Ni-Fe alloy. They are mounted in such a way that the proton beam passes through the centre of the assembly, as shown in fig. 1, thus forming a single turn primary winding.

Toroid T_1 is the main current sensor and it carries two windings which close the feedback loop of the operational amplifier OP1. This amplifier maintains a perfect balance between primary beam current (single turn) and feedback current, multiplied by n , the number of feedback turns on T_1 . This is, of course, only true for the ac components in the beam current signal, but the frequency response of this part of the circuit alone may easily descend well below 0.01 Hz.

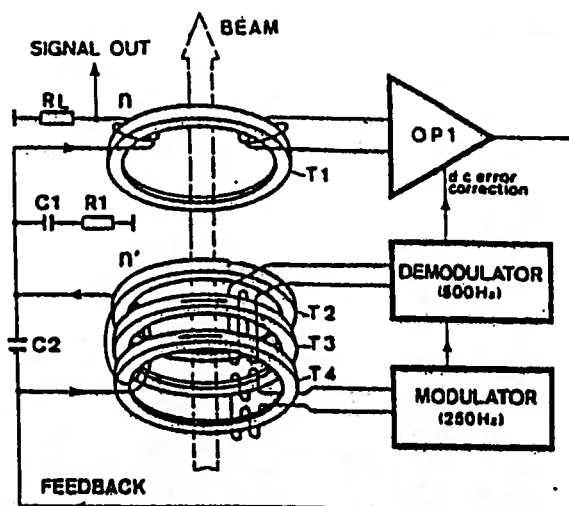


Fig. 1 - Basic schematic

The dc component of the beam current signal, or more precisely, any error in balance between effective beam current and actual feedback current is detected with a magnetic modulator-demodulator circuit ($T_2...T_4$) which produces an error correcting signal for OP1 and, therefore, the main operational loop.

The modulator pair T_3 and T_4 each carry an identical excitation winding, but they are wound in opposite directions. These two cores are driven far into saturation with the modulator frequency (250 Hz). A series winding common to both cores picks up the 2nd harmonic signal which is proportional to the effective d.c. error, for further processing in the demodulator.

Inevitable matching errors in the magnetic parameters of T_3 and T_4 will also induce modulator frequency noise in the feedback winding n' and as a result, produce a modulator frequency ripple in the output signal of the monitor. This effect is greatly reduced with the help of decoupling core T_2 (which increases leakage inductance in the coupling between modulator and feedback windings) and the short circuit capacitor C_2 .

The output signal voltage, proportional to the beam current, is derived from the feedback current flowing in precision resistor R_L .

The highest frequency components in the beam current signal do not necessarily pass via the operational amplifier and the feedback loop. The termination elements R_1 and C_1 provide a direct (passive) current return path to the load R_L . The cross-over from passive to active response of the system is linear and automatic, provided R_L and R_1 are of low value. Ideally they should be identical to the termination impedance of the coaxial transmission line which is used for signal connection. All additional loading of T_1 must be avoided, in particular the (reactive) impedance of the inter-connection to amplifier OP1 must be sufficiently high (compared with R_L).

Taking these precautions, the upper frequency limit of the combined system may be 50 MHz or more, depending only on the passive response characteristics of T_1 .

The combination of an active current transformer system (L/R integrator) with the magnetic modulator-demodulator circuit is absolutely essential, even if good high frequency performance is not required in certain applications. The L/R integrator protects the magnetic modulator from transient changes beyond its rather limited linear range, prevents hysteresis errors and rejects modulator noise either superimposed on the error correcting signal or induced in the feedback path, without introducing undesirable phase shift. This permits a high loop gain in the system (in excess of 180 dB at dc) which in turn guarantees high accuracy and linearity as a dc transformer. The ratio between primary beam and secondary feedback current is very accurately defined by n , the number of feedback turns. In practice, any ratio error at dc is less than 0.1 ppm (part per million).

The overall accuracy as a current measuring device depends ultimately on the stability and precision of the load resistor R_L . In the actual monitor system this resistor is composed of a number of precision resistors having a combined temperature coefficient of less than ± 1 ppm/ $^{\circ}\text{C}$. Each individual resistor dissipates less than 60 mW at full scale reading thus limiting the temperature rise.

In order to obtain a high degree of immunity to electromagnetic interference (EMI) in the accelerator environment, all signal and feedback circuits in the actual monitor are strictly symmetrical, for example, there are two feedback windings on each transformer feeding into two load resistors R_L in a differential mode.

The Modulator

The modulator has to provide an excitation current of constant amplitude and frequency (250 Hz) for the modulator cores and the (500 Hz) phase reference signals, for the demodulator. All frequencies are obtained by division from a quartz controlled clock.

A band pass filter and a bridge connected power amplifier provide the sinusoidal drive which is almost free from second harmonic distortion. A control loop maintains a constant peak value of excitation current, independently of load impedance.

The Demodulator

The demodulator has to detect phase and amplitude of the 2nd harmonic signal output from the modulator cores. This signal is buried in a much higher level of unwanted frequency components (modulator noise) constituted by the fundamental drive frequency and its harmonics. The origin of this "modulator noise" is the mismatch caused by the residual difference in magnetic parameters of the modulator core pair.

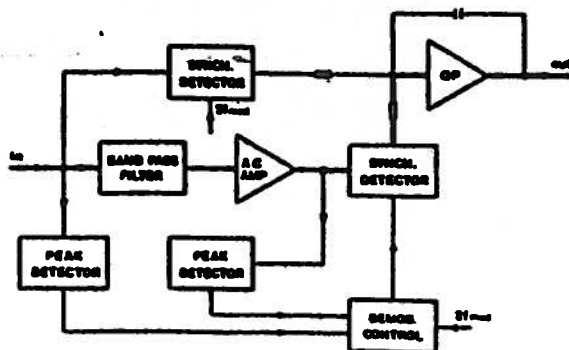


Fig. 2 - Demodulator

Shown in a simplified block diagram, fig. 2, the demodulator has two parallel channels for signal processing. The first is a synchronous detector designed for large amplitude signals and covering the full frequency range. The second channel consists of a selective, narrow band amplifier (band pass filter, ac amp.) followed again by a synchronous detector. Both channels feed into the same summing point of an active integrator with operational amplifier OP.

The narrow band amplifier is optimized for signal levels in the normal, balanced state of operation of the modulator (i.e. no dc error). This provides the high resolution in signal recovery which is necessary for minimum zero drift and low frequency noise in the monitor application. Due to higher gain, only this channel is active under normal operating conditions.

A large dc error (for example immediately after switching on), could easily saturate the high gain channel and may cause a latch-up of the monitor output. The input or output overload states of the amplifier are therefore supervised by two peak detectors which will inhibit the synchronous detector output of this channel via the demodulator control circuit. The first channel, in this case, takes over control and reduces the dc error in a short time to a value low enough to allow the high gain channel to be reactivated again.

The magnetic cores

All transformers in the system use the same type of core material - Ultraperm 10* - a high permeability Ni-Fe alloy. The core construction is a strip wound toroid (lamination thickness 50 μm) embedded without mechanical constraint in a reinforced epoxy case.

Special precautions in the manufacturing process were arranged with the manufacturer (Vacuumschmelze AG, Hanau, W. Germany) to ensure a high product quality, especially with regards to reliable insulation between magnetic layers and a low spread in magnetic parameters.

A computerized test system was made to obtain repeatable precision measurements of relevant magnetic parameters for each core. Data was taken of the static B/H curve and of the permeability μ as a function of frequency. Fifty cores of a single production batch were tested in this way and more than 200 data points were stored for each individual core on magnetic tape. From this data identical core pairs were selected for the magnetic modulator application.

The ISR Beam Current Monitor

The first application of the combined dc current transformer principle in 1970 was the circulating beam current monitor in the CERN Intersecting Storage Rings (ISR). An improved version, introduced in 1977, has the following performance:

Beam current range	-15A... 0...+80A
Resolution (200ms integration time)	$\pm 10 \text{ nA}$
Linearity	$\pm 0.001\%$
Zero drift (24 h)	$\pm 50 \text{ nA}$
Long term stability (full scale)	$\pm 0.0005\%$
Residual modulator ripple	10mA rms
Frequency range	dc .. 50MHz
Resolution of digital display	7 decimal digits
Free diameter for beam passage	150 mm

*Trade mark, Vacuumschmelze AG

High resolution systems

The antiproton projects in CERN required beam monitor systems with still higher resolution. Detailed improvements in modulator and demodulator circuit design, together with better fabrication and selection methods for the modulator cores resulted in a reduced dc drift and low frequency noise figures. As the methods previously described to eliminate residual modulator ripple in the output signal were reaching a practical limit, a second L/R integrator loop was added as shown in Fig. 3.

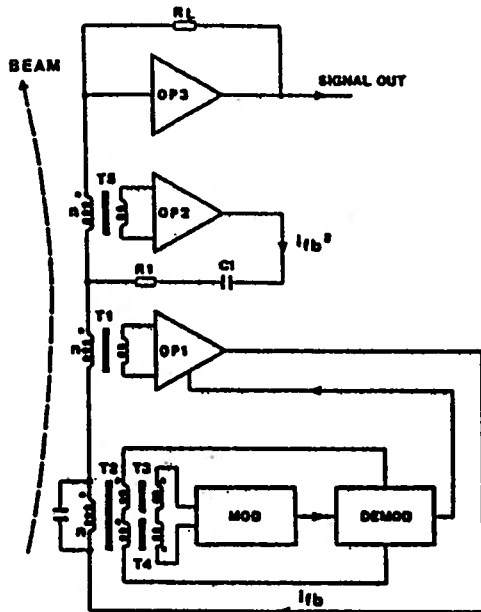


Fig. 3 - High resolution system

Modulator and demodulator, together with the first L/R integrator (T_1 , OP_1) are in the configuration discussed previously. The feedback current now passes through the feedback winding n of an additional toroid transformer T_2 . Any remaining difference between this current and the effective beam current (for example super-imposed modulator ripple) is sensed by operational amplifier OP_2 and compensated with a current $i_{fb'}$ via R_1 and C_1 .

Dynamic interaction between both feedback loops is avoided by making the R_2 a virtual zero impedance load (R_2 is in the feedback loop of operational amplifier OP_3). The dynamic characteristics of amplifier OP_3 will limit the high frequency response of this system.

The circulating beam current monitors in the CERN antiproton accumulator ring (AA) and in the CERN proton synchrotron (PS) represent this new design and satisfy the following specifications:

Beam Current Range (AA) : -500mA ... 0 ... +500mA
- 50mA ... 0 ... + 50mA*

Beam Current Range (PS) : - 2A ... 0 ... + 2A
- 20mA ... 0 ... + 20mA*

*Change of range obtained by switching the value of Load Resistance R_L

Resolution	(200 ms integration time, any range)	1 μ A
Linearity		$\pm 0.001\%$
Zero drift (24 h)		$\pm 3 \mu$ A
Long term stability (full scale)		$\pm 0.0005\%$ $\pm$ (zero drift)
Residual modulator ripple		5 μ A rms
Frequency range		dc ... 50 kHz

Mechanical Design

The ISR and the AA beam current monitor form an integral part of an ultra-high vacuum system. They contain a stainless steel vacuum chamber with a ceramic insulation gap and heating elements for a bake-out at 300°C. Fig. 4 shows the AA beam transformer assembly. The main feature of this construction (Designer : A. Maurer, CERN) is a cantilever design, where all the mechanical supporting functions are linked to one single vacuum flange. This greatly facilitates mechanical and electrical assembly in the laboratory. Concentric on the outside of the cylindrical vacuum chamber (ID = 100 mm) is a heating element, a thermal insulation, and a water-cooled thermal screen to protect the toroids from excessive temperature rise during bake-out. Heat flow between the heated flange and the external support structure is reduced using as an interconnection four stainless steel elements designed to have a low thermal conductance.

The toroids are supported by four aluminium bars which are at the same time an electrical by-pass for wall currents in the vacuum chamber system. Circular, metalised discs of fibreglass reinforced epoxy board provide individually screened compartments for each core. The electrical connections pass through thin-walled brass tubes which are soldered to the metallisation of the screens and link the windings to two external connector boxes.

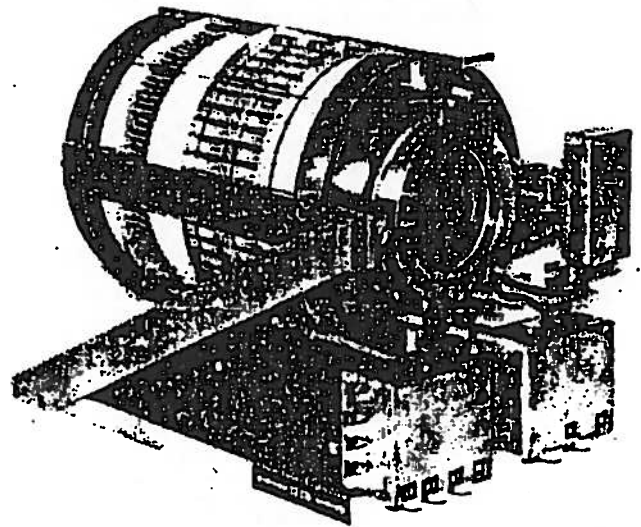


Fig. 4 - Transformer Assembly (magnetic shield removed)

References

1. J.B. Sharp, "The Induction Type Beam Current Monitor for the PS" CERN HPS-CO/62-15.
2. K. Unser, "The Circulating Beam Current Monitor" CERN-HPS-CO/68-1.
3. K. Unser, "Beam Current Transformer with DC to 300 MHz Range", IEEE Trans. Nucl. Sci., June 1969.

Instruction Manual



KEITHLEY

SPECIFICATIONS

RANGE	MAXIMUM OUTPUT	ACCURACY (1 YEAR)	STEP SIZE	TEMPERATURE COEFFICIENT
		$\pm(\%rdg + offset)$ 18°-28°C		$\pm(\%rdg + offset)/^{\circ}C$ 0°-18°C & 28°-50°C
100mA	$\pm 101.00mA$	0.1 % + 50 μA	50 μA	0.01 % + 2 μA
10mA	$\pm 19.995mA$	0.05 % + 10 μA	5 μA	0.005% + 200nA
1mA	$\pm 1.9995mA$	0.05% + 1 μA	500nA	0.005% + 20nA
100 μA	$\pm 199.95 \mu A$	0.05% + 100nA	50nA	0.005% + 2nA
10 μA	$\pm 19.995 \mu A$	0.05% + 10nA	5nA	0.005% + 200pA

OUTPUT RESISTANCE: Greater than $10^{12}\Omega$.

OUTPUT CAPACITANCE: Less than 20pF.

LINE REGULATION: Less than 0.01% for AC power line changes within specified limits.

VOLTAGE LIMIT: Bipolar, 1 to 105V in 1V programmable steps.

RESPONSE TIME: Less than 3ms to within 0.1% of programmed change.

TRANSIENT RECOVERY TIME: Less than 3ms to rated accuracy following any change in compliance voltage.

NOISE

RANGE	NOISE (p-p of range)	BANDWIDTH
100mA	100ppm	0.1Hz to 30kHz
10 μA to 10mA	100ppm	0.1Hz to 100Hz

GUARD OUTPUT: Maximum Load Capacitance: 10nF.

Maximum Load Current: Absolute total (Output + Guard) not to exceed 105mA.

Accuracy: $\pm 1mV$ (excluding output lead voltage drop).

INCREMENT/DECREMENT: Automatic, manual or trigger modes.

Range of Dwell Times: 50ms to 999.9s.

Accuracy of Dwell Times: $\pm(0.05\% + 20\mu s)$.

Step Size: Selected digit on a fixed range. Minimum step size 0.1% of range.

Current Limit: Maximum is $\pm$ (Full Scale) on range selected.

OUTPUT LOAD: Output load must be non-inductive

EXTERNAL TRIGGER: TTL-compatible EXTERNAL TRIGGER INPUT and OUTPUT.

OUTPUT CONNECTIONS: Teflon® insulated triax connector (Specialty Connector #30JR121-1) for output; five way binding posts for GUARD OUTPUT COMMON and CHASSIS; BNC (chassis isolated) connectors for EXTERNAL TRIGGER INPUT and OUTPUT. All connections on rear panel.

IEEE-488 BUS INTERFACE (option 2243)

Multiline Commands: DCL, LLO, SDC, GET, GTL, UNT, UNL, SPE, SPD.

Uniline Commands: IFC, REN, EOI, SRQ, ATN.

Interface Functions: SH1, AH1, T6, TED, L4, LED, SR1, RL1, PP0, DC1, DT1, C0, E1.

Output Connections: Amphenol or Cinch Series 87 IEEE and printed circuit digital I/O port. All connections on rear panel.

Internal Programmable Parameters: Display Mode, Output, (Prefix Data Format), EOI, SRQ (including mask for over V-limit), Range, Terminator Character, Inputs (SOURCE, V-LIMIT, DWELL TIME), Output Status.

Digital I/O Port: A separate I/O port consisting of four input and four output lines as well as common (IEEE-488) and +5VDC. Outputs will drive one TTL load. Inputs represent one TTL load. The 224 can be programmed to generate a "SRQ" upon any change in the four bit input data. Mating connector supplied.

GENERAL

DISPLAY: 0.5" LED digits, 4 1/2-digit signed mantissa, 1-digit signed exponent.

OVER VOLTAGE LIMIT INDICATION: "V-LIMIT" LED will blink.

SELF TEST: Digital RAM, ROM, and front panel LEDs upon power up.

WARMUP: 1 hour to rated accuracy.

POWER: 105-125 or 210-250VAC (internal switch selected), 50 or 60Hz, 60 watts maximum (80VA maximum). 90-105 or 180-210VAC operation available.

COOLING: Internal fan for forced air cooling.

ENVIRONMENTAL LIMITS: Operating: 0°-50°C; up to 35°C at 70% non-condensing relative humidity. Storage: -25°C to 70°C.

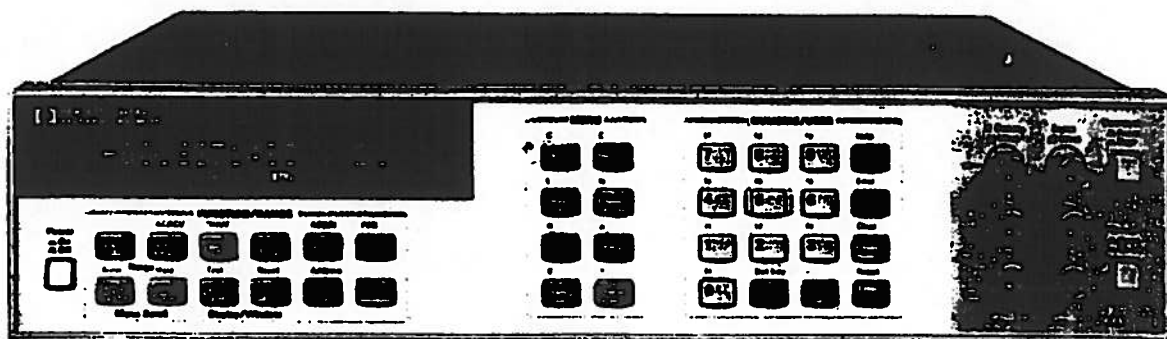
DIMENSIONS, WEIGHT: 127mm high x 216mm wide x 359mm deep (5-1/2" x 8-1/2" x 14-1/8"). Net weight 9 lbs. 15 oz. (4.52kg).

MAXIMUM ALLOWABLE COMMON MODE VOLTAGE (OUTPUT or OUTPUT COMMON to CHASSIS): 250V rms, DC to 60Hz.

ACCESSORIES SUPPLIED: Model 6011: Triaxial Test Lead (3 ft.)

DIGITAL MULTIMETERS/DIGITAL VOLTMETERS

System Multimeter
HP 3458A



HP 3458A



The HP 3458A multimeter shatters longstanding performance barriers of speed and accuracy on the production test floor, in research and development, and in the calibration lab. The HP 3458A is the fastest, most flexible, and most accurate multimeter offered by Hewlett-Packard. In your system or on the bench, the HP 3458A saves you time and money with unprecedented test system throughput and accuracy, seven-function measurement flexibility, and low cost of ownership.

Select a rate of 100,000 readings per second for maximal test throughput. Or achieve highest levels of precision with up to 8½ digits of measurement resolution and 0.1 part per million transfer accuracy. Add to this the HP 3458A's simplicity of operation, and you have the ideal multimeter for your most demanding applications.

High-Test System Throughput

Faster Testing

- Up to 100,000 readings/s
- Internal test setups >340/s
- Programmable integration times from 500 ns to 1 s

Greater Test Yield

- More accuracy for tighter test margins
- Up to 8½ digits resolution

Longer Uptime

- Two-source (10 V, 100 kΩ) calibration, including ac
- Self-adjusting, self-verifying autocalibration for all functions and ranges, including ac

High-Resolution Digitizing

Greater Waveform Resolution and Accuracy

- 16 to 24 bits resolution
- 100,000 to 0.2 samples/s
- 12 MHz bandwidth
- Timing resolution to 10 ns
- Less than 100 ps time jitter
- Over 75,000 reading internal memory

Calibration Lab Precision

Superb Transfer Measurements

- 8½ digits resolution
- 0.1 ppm dc volts linearity
- 0.1 ppm dc volts transfer capability
- 0.01 ppm rms internal noise

Extraordinary Accuracy

- 0.6 ppm for 24 hours in dc volts
- 2.2 ppm for 24 hours in Ω
- 100 ppm mid-band ac volts
- 8 ppm (4 ppm optional) per year voltage reference stability

HP 3458A Multimeter Performance Features

DC Volts

- 5 ranges: 0.1 V to 1000 V
- 8½ to 4½ digits resolution
- Up to 100,000 readings/s (4½ digits)
- Maximum sensitivity: 10 nV
- 0.6 ppm 24-hour accuracy
- 8 ppm (4 ppm optional)/year voltage reference stability

Ω

- 9 ranges: 10 Ω to 1 GΩ
- 2-wire and 4-wire Ω with offset compensation
- Up to 50,000 readings/second (5½ digits)
- Maximum sensitivity: 10 μΩ
- 2.2 ppm 24-hour accuracy

AC Volts

- 6 ranges: 10 mV to 1000 V
- 1 Hz to 10 MHz bandwidth
- Up to 50 readings/s with all readings to specified accuracy
- Choice of sampling or analog true rms techniques
- 100 ppm best accuracy

DC Current

- 8 ranges: 100 nA to 1 A
- Up to 1,350 readings/s (5½ digits)
- Maximum sensitivity: 1 pA
- 14 ppm 24-hour accuracy

AC Current

- 5 ranges: 100 μA to 1 A
- 10 Hz to 100 kHz bandwidth
- Up to 50 readings/second
- 500 ppm 24-hour accuracy

Frequency and Period

- Voltage or current ranges
- Frequency: 1 Hz to 10 MHz
- Period: 100 ns to 1 second
- 0.01 % accuracy
- AC or dc coupled

4/AC Voltage

General Information

The HP3458A supports three techniques for measuring true rms AC voltage, each offering unique capabilities. The desired measurement technique is selected through the SETACV command. The ACV functions will then apply the chosen method for subsequent measurements.

The following section provides a brief description of the three operation modes along with a summary table helpful in choosing the technique best suited to your specific measurement need.

SETACV SYNC Synchronously Sub-sampled Computed true rms technique.

This technique provides excellent linearity and the most accurate measurement results. It does require that the input signal be repetitive (not random noise for example). The bandwidth in this mode is from 1 Hz to 10 MHz.

SETACV ANA Analog Computing true rms conversion technique.

This is the measurement type at power-up or following an instrument reset. This mode works well with any signal within its 10 Hz to 2 MHz bandwidth and provides the fastest measurement speeds.

SETACV RNDM Random Sampled Computed true rms technique.

This technique again provides excellent linearity, however the overall accuracy is the lowest of the three modes. It does not require a repetitive input signal and is therefore well suited to wideband noise measurements. The bandwidth in this mode is from 20 Hz to 10 MHz.

Selection Table

Technique	Frequency Range	Best Accuracy	Repetitive Signal Required	Readings / Sec.	
				Minimum	Maximum
Synchronous Sub-sampled	1 Hz - 10 MHz	0.010%	Yes	0.025	10
Analog	10 Hz - 2 MHz	0.03%	No	0.8	50
Random Sampled	20 Hz - 10 MHz	0.1%	No	0.025	45

Synchronously Sub-sampled Mode (ACV Function, SETACV SYNC)

Range	Full Scale	Maximum Resolution ± 1 ppm	Input Impedance	Temperature Coefficient ¹ (% of Reading + % of Range) / °C
10 mV	12.00000	10 nV	1 MΩ ± 15% with < 140 pF	0.002 + 0.02
100 mV	120.0000	100 nV	1 MΩ ± 15% with < 140 pF	0.001 + 0.0001
1 V	1.200000	1 μV	1 MΩ ± 15% with < 140 pF	0.001 + 0.0001
10 V	12.00000	10 μV	1 MΩ ± 2% with < 140 pF	0.001 + 0.0001
100 V	120.0000	100 μV	1 MΩ ± 2% with < 140 pF	0.001 + 0.0001
1000 V	1050.000	1 mV	1 MΩ ± 2% with < 140 pF	0.001 + 0.0001

¹ Additional error beyond ± 1°C, but within ± 5°C of last ACAL.

For ACBAND > 2 MHz, use 10 mV range temperature coefficient for all ranges.

² Specifications apply full scale to 10% of full scale, DC < 10% of AC, sine wave input, crest factor = 1.4, and PRESET. Within 24 hours and ± 1°C of last ACAL.

Peak (AC + DC) input limited to 5 x full scale for all ranges in ACV function.

Add 2 ppm of reading additional error for HP factory traceability of 10 V DC to US NBS.

AC Accuracy²

24 Hour to 2 Year (% of Reading + % of Range)

Range	ACBAND ≤ 2 MHz							
	1 Hz to 40 Hz	40 Hz to 1 kHz	1 kHz to 20 kHz	20 kHz to 60 kHz	60 kHz to 100 kHz	100 kHz to 300 kHz	300 kHz to 1 MHz	1 MHz to 2 MHz
10 mV	0.03 + 0.03	0.02 + 0.011	0.03 + 0.011	0.1 + 0.011	0.5 + 0.011	4.0 + 0.02		
100 mV - 10 V	0.007 + 0.004	0.007 + 0.002	0.014 + 0.002	0.03 + 0.002	0.08 + 0.002	0.3 + 0.01	1 + 0.01	1.5 + 0.01
100 V	0.02 + 0.004	0.02 + 0.002	0.02 + 0.002	0.035 + 0.002	0.12 + 0.002	0.4 + 0.01	1.5 + 0.01	
1000 V	0.04 + 0.004	0.04 + 0.002	0.06 + 0.002	0.12 + 0.002	0.3 + 0.002			

AC Accuracy continued on following page.

AC Accuracy continued: 24 Hour to 2 Year (% of Reading + % of Range)

Range	ACBAND > 2 MHz				
	45 Hz to 100 kHz	100 kHz to 1 MHz	1 MHz to 4 MHz	4 MHz to 8 MHz	8 MHz to 10 MHz
10 mV	0.09 + 0.06	1.2 + 0.05	3 + 0.07	20 + 0.08	
100 mV - 10 V	0.09 + 0.06	2.0 + 0.05	4 + 0.07	4 + 0.08	15 + 0.1
100 V	0.12 + 0.002				
1000 V	0.3 + 0.01				

Transfer Accuracy

Range	% of Reading	Conditions
100 mV - 100 V	(0.002 + Resolution in %) ¹	• Following 4 Hour warm-up • Within 10 Min and $\pm 0.5^{\circ}\text{C}$ of the reference measurement • 45 Hz to 20 kHz, sine wave input • Within $\pm 10\%$ of the reference voltage and frequency

¹ Resolution is the value of RES command or parameter.

² Additional error beyond $\pm 1^{\circ}\text{C}$, but within $\pm 5^{\circ}\text{C}$ of last ACAL (% of Range) / $^{\circ}\text{C}$. For ACBAND > 2 MHz, use 10 mV range temperature coefficient.

AC + DC Accuracy (ACDCV Function)

For ACDCV Accuracy apply the following additional error to the ACV accuracy. (% of Range)

DC < 10% of AC Voltage

Range	ACBAND $\leq$ 2 MHz	ACBAND > 2 MHz	Temperature Coefficient ²
10 mV	0.09	0.09	0.03
100 mV - 1000 V	0.008	0.09	0.0025

DC > 10% of AC Voltage

Range	ACBAND $\leq$ 2 MHz	ACBAND > 2 MHz	Temperature Coefficient ²
10 mV	0.7	0.7	0.18
100 mV - 1000 V	0.07	0.7	0.025

Additional Errors

Apply the following additional errors as appropriate to your particular measurement setup. (% of Reading)

Source R	Input Frequency ³				Crest Factor	Resolution Multiplier ¹
	0 - 1 MHz	1 - 4 MHz	4 - 8 MHz	8 - 10 MHz		
0 Ω	0	2	5	5	1 - 2	(Resolution in %) x 1
50 Ω Termination	0.003	0	0	0	2 - 3	(Resolution in %) x 2
75 Ω Termination	0.004	2	5	5	3 - 4	(Resolution in %) x 3
50 Ω	0.005	3	7	10	4 - 5	(Resolution in %) x 5

³ Flatness error including instrument loading.

Reading Rates ⁴

ACBAND Low	Maximum Sec / Reading
1 - 5 Hz	6.5
5 - 20 Hz	2.0
20 - 100 Hz	1.2
100 - 500 Hz	0.32
> 500 Hz	0.02

% Resolution	Maximum Sec / Reading
0.001 - 0.005	32
0.005 - 0.01	6.5
0.01 - 0.05	3.2
0.05 - 0.1	0.64
0.1 - 1	0.32
> 1	0.1

⁴ Reading time is the sum of the Sec / Reading shown for your configuration. The tables will yield the slowest reading rate for your configuration. Actual reading rates may be faster. For DELAY-1; ARRANGE OFF.

Settling Characteristic

There is no instrument settling required.

Common Mode Rejection

For 1 k Ω imbalance in LO lead, > 90 dB, DC to 60 Hz.

High Frequency Temperature Coefficient

For outside Tcal $\pm 5^{\circ}\text{C}$ add the following error.
(% of Reading) / $^{\circ}\text{C}$

Range	Frequency	
	2 - 4 MHz	4 - 10 MHz
10 mV - 1 V	0.02	0.08
10 V - 1000 V	0.08	0.08

Maximum Input

	Rated Input	Non-Destructive
HI to LO	$\pm 1000 \text{ V pk}$	$\pm 1200 \text{ V pk}$
LO to Guard	$\pm 200 \text{ V pk}$	$\pm 350 \text{ V pk}$
Guard to Earth	$\pm 500 \text{ V pk}$	$\pm 1000 \text{ V pk}$
Volt - Hz Product	1×10^8	

Analog Mode (ACV Function, SETACV ANA)

Range	Full Scale	Maximum Resolution	Input Impedance	Temperature Coefficient ¹ (% of Reading + % of Range) / $^{\circ}\text{C}$
10 mV	12.00000	10 nV	1 M Ω $\pm$ 15% with < 140 pF	0.003 + 0.006
100 mV	120.0000	100 nV	1 M Ω $\pm$ 15% with < 140 pF	0.002 + 0.0
1 V	1.200000	1 μV	1 M Ω $\pm$ 15% with < 140 pF	0.002 + 0.0
10 V	12.00000	10 μV	1 M Ω $\pm$ 2% with < 140 pF	0.002 + 0.0
100 V	120.0000	100 μV	1 M Ω $\pm$ 2% with < 140 pF	0.002 + 0.0
1000 V	1050.000	1 mV	1 M Ω $\pm$ 2% with < 140 pF	0.002 + 0.0

¹ Additional error beyond $\pm 1^{\circ}\text{C}$, but within $\pm 5^{\circ}\text{C}$ of last ACAL.

² Specifications apply full scale to 1/20 full scale, sine wave input, crest factor = 1.4, and PRESET. Within 24 hours and $\pm 1^{\circ}\text{C}$ of last ACAL.

Maximum DC is limited to 400 V in ACV function.

Add 2 ppm of reading additional error for HP factory traceability of 10 V DC to US NBS.

AC Accuracy²

24 Hour to 2 Year (% Reading + % Range)

Range	10 Hz to 20 Hz	20 Hz to 40 Hz	40 Hz to 100 Hz	100 Hz to 20 kHz	20 kHz to 50 kHz	50 kHz to 100 kHz	100 kHz to 250 kHz	250 kHz to 500 kHz	500 kHz to 1 MHz	1 MHz to 2 MHz
10 mV	0.4 + 0.32	0.15 + 0.25	0.06 + 0.25	0.02 + 0.25	0.15 + 0.25	0.7 + 0.35	4 + 0.7			
100 mV - 10 V	0.4 + 0.02	0.15 + 0.02	0.06 + 0.01	0.02 + 0.01	0.15 + 0.04	0.6 + 0.08	2 + 0.5	3 + 0.6	5 + 2	10 + 5
100 V	0.4 + 0.02	0.15 + 0.02	0.06 + 0.01	0.03 + 0.01	0.15 + 0.04	0.5 + 0.08	2 + 0.5	3 + 0.6	5 + 2	
1000 V	0.42 + 0.03	0.17 + 0.03	0.08 + 0.02	0.06 + 0.02	0.15 + 0.04	0.6 + 0.2				

AC + DC Accuracy (ACDCV Function)

For ACDCV Accuracy apply the following additional error to the ACV accuracy. (% of Reading + % of Range)

Range	DC < 10% of AC Voltage		DC > 10% of AC Voltage	
	Accuracy	Temperature Coefficient ³	Accuracy	Temperature Coefficient ³
10 mV	0.0 + 0.2	0 + 0.015	0.15 + 3	0 + 0.06
100 mV - 1000 V	0.0 + 0.02	0 + 0.001	0.15 + 0.25	0 + 0.007

³ Additional error beyond $\pm 1^{\circ}\text{C}$, but within $\pm 5^{\circ}\text{C}$ of last ACAL. (% of Reading + % of Range) / $^{\circ}\text{C}$.

Additional Errors

Apply the following additional errors as appropriate to your particular measurement setup.

Low Frequency Error (% of Reading)

Signal Frequency	ACBAND Low		
	10 Hz - 1 kHz NPLC > 10	1 - 10 kHz NPLC > 1	> 10 kHz NPLC > 0.1
200 - 500 Hz	0	0.15	
500 - 1 kHz	0	0.015	0.9
1 - 2 kHz	0	0	0.2
2 - 5 kHz	0	0	0.05
5 - 10 kHz	0	0	0.01

Crest Factor Error (% of Reading)

Crest Factor	Additional Error
1 - 2	0
2 - 3	0.15
3 - 4	0.25
4 - 5	0.40

Reading Rates¹

ACBAND Low	Sec / Reading		
	NPLC	ACV	ACDCV
≥ 10 Hz	10	1.2	1
≥ 1 kHz	1	1	0.1
≥ 10 kHz	0.1	1	0.02

¹ For DELAY-1; ARANGE OFF
For DELAY 0; NPLC .1, uns.
reading rates of greater than 500 /
Sec are possible.

Settling Characteristic

For first reading or range change error using default delays, add .01% of input step additional error.
The following data applies for DELAY 0.

Function	ACBAND Low	DC Component	Settling Time	†
ACV	≥ 10 Hz	DC < 10% AC	0.5 sec to 0.01%	
		DC > 10% AC	0.9 sec to 0.01%	
ACDCV	10 Hz - 1 kHz		0.5 sec to 0.01%	
	1 kHz - 10 kHz		0.08 sec to 0.01%	
	≥ 10 kHz		0.015 sec to 0.01%	

Maximum Input

	Rated Input	Non-Destructive
HI to LO	± 1000 V pk	± 1200 V pk
LO to Guard	± 200 V pk	± 350 V pk
Guard to Earth	± 500 V pk	± 1000 V pk
Volt - Hz Product	1 x 10 ⁸	

Common Mode Rejection

For 1 kΩ imbalance in LO lead, >90 dB, DC - 60 Hz.

Random Sampled Mode (ACV Function, SETACV RNDM)

Range	Full Scale	Maximum Resolution	Input Impedance	Temperature Coefficient ² (% of Reading + % of Range) / °C
10 mV	120.0000	10 nV	1 MΩ ± 15% with < 140 pF	0.002 + 0.02
100 mV	120.0000	100 nV	1 MΩ ± 15% with < 140 pF	0.001 + 0.0001
1 V	1.200000	1 μV	1 MΩ ± 15% with < 140 pF	0.001 + 0.0001
10 V	12.00000	10 μV	1 MΩ ± 2% with < 140 pF	0.001 + 0.0001
100 V	120.0000	100 μV	1 MΩ ± 2% with < 140 pF	0.001 + 0.0001
1000 V	1050.000	1 mV	1 MΩ ± 2% with < 140 pF	0.001 + 0.0001

² Additional error beyond ±1°C,
but within ±5°C of last ACAL.
For ACBAND > 2 MHz, use
10 mV range temperature
coefficient for all ranges.

AC Accuracy³

24 Hour to 2 Year (% of Reading + % of Range)

Range	ACBAND ≤ 2 MHz				ACBAND > 2 MHz				
	20 Hz to 100 kHz	100 kHz to 300 kHz	300 kHz to 1 MHz	1 MHz to 2 MHz	20 Hz to 100 kHz	100 kHz to 1 MHz	1 MHz to 4 MHz	4 MHz to 8 MHz	8 MHz to 10 MHz
10 mV	0.5+0.02	4+0.02			0.1+0.05	1.2+0.05	3+0.07	20+0.08	
100 mV - 10 V	0.08+0.002	0.3+0.01	1+0.01	1.5+0.01	0.1+0.05	2+0.05	4+0.07	4+0.08	15+0.1
100 V	0.12+0.002	0.4+0.01	1.5+0.01		0.12+0.002				
1000 V	0.3+0.01				0.3+0.01				

³ Specifications apply from full
scale to 5% of full scale, DC
< 10% of AC, sine wave input,
crest factor = 1.4, and PRESET.
Within 24 hours and ±1°C of last
ACAL.

Add 2 ppm of reading additional
error for HP factory traceability of
10 V DC to US NBS.

Maximum DC is limited to 400 V
in ACV function.

AC + DCV Accuracy (ACDCV Function)

For ACDCV Accuracy apply the following additional error to the ACV accuracy. (% of Range).

Range	DC ≤ 10% of AC Voltage			DC > 10% of AC Voltage		
	ACBAND ≤ 2 MHz	ACBAND > 2 MHz	Temperature Coefficient ¹	ACBAND ≤ 2 MHz	ACBAND > 2 MHz	Temperature Coefficient ¹
10 mV	0.09	0.09	0.03	0.7	0.7	0.18
100 mV - 1 kV	0.008	0.09	0.0025	0.07	0.7	0.025

Additional Errors

Apply the following additional errors as appropriate to your particular measurement setup. (% of Reading)

Source R	Input Frequency ²				Crest Factor	Resolution Multiplier
	0 - 1 MHz	1 - 4 MHz	4 - 8 MHz	8 - 10 MHz		
0 Ω	0	2	5	5	1 - 2	(Resolution in %) x 1
50 Ω Termination	0.003	0	0	0	2 - 3	(Resolution in %) x 3
75 Ω Termination	0.004	2	5	5	3 - 4	(Resolution in %) x 5
50 Ω	0.005	3	7	10	4 - 5	(Resolution in %) x 8

¹ Additional error beyond ±1°C, but within ±5°C of last ACAL (% of Reading) / °C.

For ACBAND > 2 MHz, use 10 mV range temperature coefficient for all ranges.

² Flatness error including instrument loading.

Reading Rates ³

% Resolution	Sec / Reading	
	ACV	ACDCV
0.1 - 0.2	40	39
0.2 - 0.4	11	9.6
0.4 - 0.6	2.7	2.4
0.6 - 1	1.4	1.1
1 - 2	0.8	0.5
2 - 5	0.4	0.1
> 5	0.32	0.022

High Frequency Temperature Coefficient

For outside Tcal ±5°C add the following error. (% of Reading) / °C

Range	2 - 4 MHz	4 - 10 MHz
10 mV - 1V	0.02	0.08
10 V - 1000 V	0.08	0.08

³ For DELAY -1; RANGE OFF. For DELAY 0 in ACV, the reading rates are identical to ACDCV.

Settling Characteristic

For first reading or range change error using default delays, add 0.01% of input step additional error. The following data applies for DELAY 0.

Function	DC Component	Settling Time
ACV	DC < 10% of AC	0.5 sec to 0.01%
	DC > 10% of AC	0.9 sec to 0.01%
ACDCV	No instrument settling required.	

Common Mode Rejection

For 1 kΩ imbalance in LO lead, >90 dB, DC to 60 Hz.

Maximum Input

	Rated Input	Non-Destructive
HI to LO	± 1000 V pk	± 1200 V pk
LO to Guard	± 200 V pk	± 350 V pk
Guard to Earth	± 500 V pk	± 1000 V pk
Volt - Hz Product	1 x 10 ⁸	

11/General Specifications

Operating Environment
0°C to 55°C

Storage Temperature
-40°C to +75°C

Operating Humidity Range
up to 95% RH at 40°C

Warm-Up Time
4 Hours to published specifications

Physical Characteristics
88.9mm H x 425.5mm W x 502.9mm D
Net Weight: 12kg (26.5 lbs)
Shipping Weight 14.8kg (32.5 lbs)

Power Requirements
100/120 V, 220/240 V $\pm$ 10%
48 Hz to 420 Hz automatically sensed
< 30 W, < 80 VA (peak)
Fused: 1.5 A @ 115 V or 0.5 A @ 230 V

IEEE-488 Interface
Complies with the following:
IEEE-488.1 Interface Standard
IEEE-728 Codes/Formats Standard
HPML (HP Multimeter Language)

Designed in Accordance with
Safety: IEC 348, UL1244, CSA
EMI: FTZ 1046, FCC part 15-J

Input Terminals
Gold-plated Tellurium Copper

Field Installation Kits		HP Part Number
Option 001	Extended Reading Memory	03458-87901
Option 002	High Stability Reference	03458-80002
Option 005	Waveform Analysis Library	03458-80005
Extra Keyboard Overlays (5 each)		03458-84303

Available Documentation	HP Part Number
Product Note 3458A-1: Optimizing Throughput and Reading Rate	5953-7058
Product Note 3458A-2: High Resolution Digitizing with the HP 3458A	5953-7059
Product Note 3458A-3: Electronic Calibration of the HP 3458A	5953-7060
Extra Operating Manual	03458-90000

Waveform Analysis Library Description

Provides a library of subprograms and functions that can initialize the computer and the HP 3458A, acquire waveform data, perform parametric measurements on the data, compare the waveform data to previously captured limits, and format and output the measurement results in a real-time scope-like fashion on the display.

Software routines are also provided for time domain analysis (frequency, rise time, pulse width) and frequency domain analysis (Fast Fourier transforms, FFT). Routines also provide interpolation between data points (linear or $\sin x / x$).

These documents will replace the corresponding pages when the 10kHz system will be operationnal, year 1998.

For more details, see John Musson.

energy W stored in the cavity and the total dissipated power P_d , weighted by the resonant frequency ω :

$$Q = \frac{\omega \cdot W}{P_d}$$

or if we assume that the dissipation of the energy occurs only on the walls of the cavity:

$$Q = \frac{\text{volume}}{\text{area}}$$

$Q=3000$ for these cavities.

Temperature dependance:

From the equations one see that Q is sensitive to the resonant frequency of the cavity and consequently to the dimensions of the cavity. Since the change of the dimensions is related to the variation of the temperature, to perform an accurate measurement one must be sure that the temperature is constant.

Position dependance:

The power coupled from the beam to the cavity at radius ρ also depends for the TM010 mode (this mode is relatively insensitive to the position of the beam) to the variations of the electric field along the z axis $E_z\rho$:

$$P = \text{const} \cdot I_{\text{beam}} \cdot E_z\rho$$

thus the variation of E_z from the change in the beam position represents the position dependance of the current measurement.

The effect of the beam displacement is about 0.06% /cm.

Performances & Limitations:

Unlike the Unser the cavities are not absolute measurements, they are only relative because you collect a signal which is proportionnal to the beam current. so they need to be calibrated using an absolute measurement like the Unser monitor.

However, as they are very stable and linear, once they have been well calibrated, that is you have transfered to them the accuracy of the Unser, you can use them for the regular measurement on all the dynamic range.

Associated Electronics

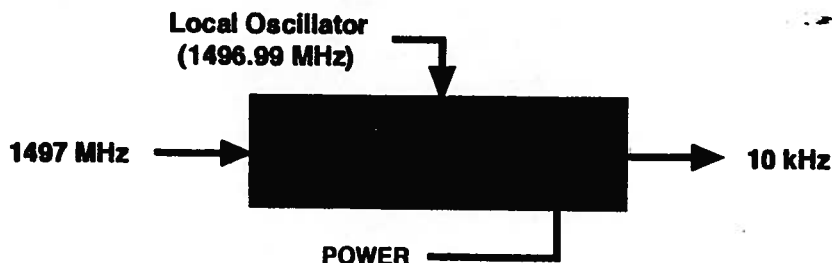
Unser electronics:

Below the grey box in the hall is located the Front-End Electronic of the Parametric Current Transformer. As it is located 1m below the beam line, this black box is shielded with lead bricks. This box needs to be removed to adjust the offset of the Unser (see Annexe 1).

The second part, called Back-End Electronic, is located in the counting house in rack #CH01B06. From there all the controls can be operated manually.

Down-Converters:

The purpose of this system is to downconvert a high-frequency signal (1497MHz RF) in an intermediate frequency signal (10kHz IF for example). The input RF signal is mixed with a local oscillator signal to get the output IF signal:



INSTRUCTIONS MANUALS

**PARAMETRIC CURRENT TRANSFORMER
DOWNCONVERTER
RMS TO DC CONVERTER**

SAFETY INSTRUCTIONS

This instrument is operated from a power source which applies up to 115 Volts rms between the supply conductors or between either supply conductor and the ground. A protective grounding by way of the grounding conductor in the power cord is essential for safe operation.

To avoid fire hazard, use only the fuse of the correct type, current rating and voltage rating as specified in the 'Preparation for use' herewith.

Do not remove cover panels while the instrument is powered. Do not operate the instrument without the cover panels properly installed.

GENERAL DESCRIPTION

The parametric-current-transformer is an instrument designed to measure the intensity of a charged particle beam over a wide frequency range including dc. It incorporates the latest available technologies in respect to magnetic material, electronics design and manufacturing techniques. It consists of an active current transformer and a magnetic parametric amplifier in a common feedback loop. It features a large dynamic range, high resolution, excellent linearity and long term stability.

Its characteristics also makes it the ideal instrument to measure the electric current in a conductor when galvanic isolation, up to very high voltages, is required.

The magnetic parametric amplifier uses a magnetic modulator of novel design. A digital excitation generator drives its modulator in an avalanche mode with high peak currents. This reduces the effect of Barkhausen noise and improves dc zero stability in a very important way.

The demodulator uses a parametric amplifying mechanism with a very good signal to noise ratio. It virtually eliminates the contribution of noise from active amplifier circuits.

The toroidal cores of the parametric transformer are made out of thin ribbons of a high permeability alloy with amorphous structure. It is a material developed recently in application of a new processing technology.

SUMMARY

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General Description	3
Theory of Operation	4
Systems Components	4
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ANNEXES

Plots of resolution vs. integration time	
Range A full bandwidth ripple signal printout	
Range B full bandwidth ripple signal printout	
Range A full bandwidth full scale square wave output printout	
Range B full bandwidth full scale square wave output printout	
The Parametric Current Transformer, a universal beam intensity monitor, by K.B. Unser	

However, the present application has specific requirements which can not be met by any commercially available magnetic core, neither of the older metallic crystalline type, nor of the chosen amorphous ribbon type.

We therefore developed a special manufacturing process to improve the magnetic characteristic stability of the parametric current transformer cores. In addition, these cores require a very careful packaging and a sophisticated annealing treatment, both thermal and magnetic, using a longitudinal field and a transverse field.

A multilayer magnetic shield using amorphous alloys provides a good shielding factor from external magnetic fields. This is important if low values of current have to be monitored.

The Parametric Current Transformer was developed by Dipl. Ing. Klaus B. Unser of the LEP Division at CERN in the framework of the Collaboration Agreement N° K017/LEP between CERN and BERGOZ. This agreement grants BERGOZ the right to commercialize the resulting technology.

THEORY OF OPERATION

The working principle of DC Beam Current Transformers is described in publications attached to this manual:

The Parametric Current Transformer, a universal beam intensity monitor
 by K. Unser

paper presented at the 1991 Accelerator Instrumentation Workshop
 CERN, Newport News VA, U.S.A., October 28-31, 1991

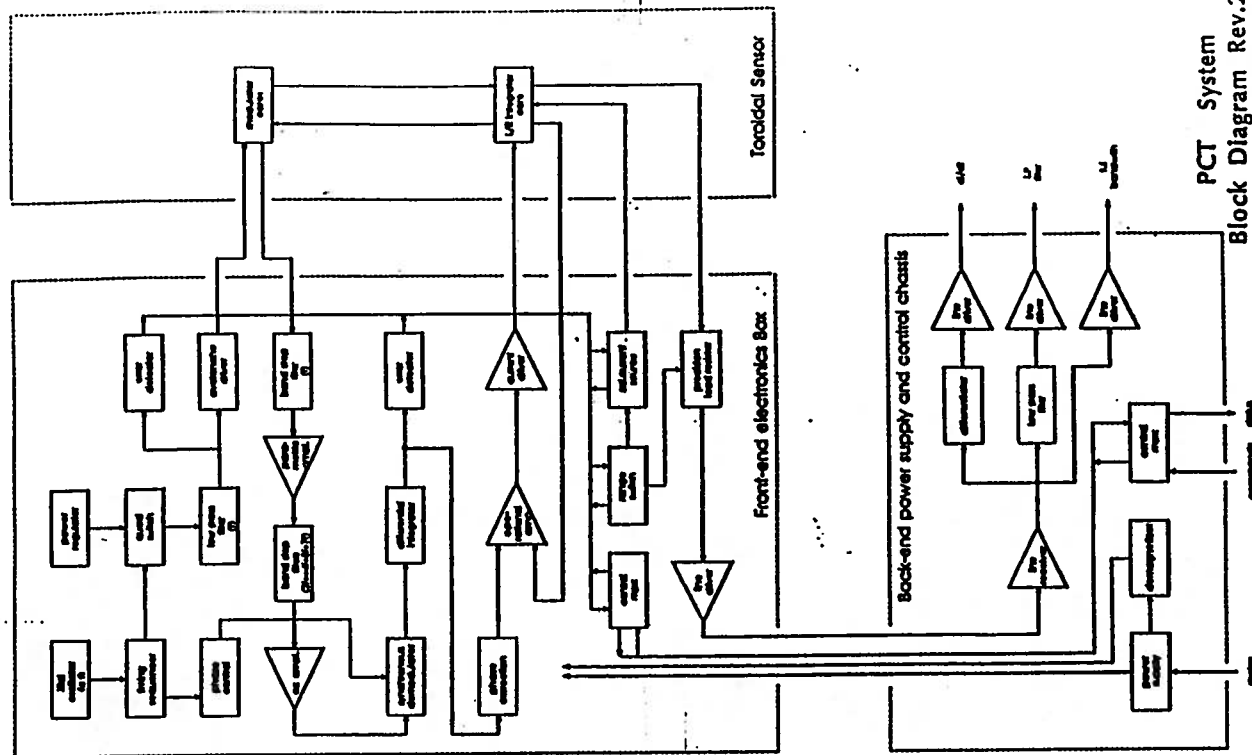
SYSTEMS COMPONENTS

The system consists of:

- Toroidal sensor
- Front-end electronics box
- Back-end power supply and control chassis

The toroidal sensor is connected to the electronics box by two multicore cables equipped with 8-pin Burydy connectors.

A multicore cable equipped with 12-pin Burydy connectors at each end connects the front-end electronics box to the back-end chassis. This cable can be up to 300 meters long.



PCT System
 Block Diagram Rev.2.0

PREPARATION FOR USE

Before connecting to the mains, connect the system's components as described above. Each connector is equipped with dummy pins which only allows the connection to which it is intended.

Connect an oscilloscope or voltmeter set for ± 10 Volts full scale to the middle BNC connector labelled 'LP-Filter'. It is located at the front of the back-end chassis.

Set the three front panel switches as follows:

- Calibration: 'Off'
- Range: 'A'
- Test: 'Off'

Check the line voltage for the unit marked at the rear of the back-end chassis. Make sure it conforms to your actual mains voltage.

The main line fuse is located behind a trap, next to the mains connector of the back-end chassis. 230-Volts units must use 1.25 A slo blo fuses 5x20mm. 100- and 115-Volts units must use 2.5 A slo blo fuses 5x20mm.

The PCT system is not equipped with a mains power switch because it is intended to remain powered permanently: the system consumes less than 10VA.

Connect the back-end chassis to the mains.

This starts a low frequency high current demagnetizing sequence which may last up to 60 seconds depending on sensor size. It is important for the performance of the instrument that this sequence is not interrupted. Whenever the power is interrupted during this sequence, the demagnetization process is incomplete and must be restarted. To do that, remove the power plug from the chassis socket, wait at least 15 minutes and reapply the mains power again.

After the demagnetizing sequence is completed, the PCT reading corresponds to the magnetic environment of the toroidal sensor, i.e. earth magnetic field and mains related fields. Rotate and displace the sensor to identify the nature of the fields indicated.

ANALOG OUTPUTS

The BNC connectors on the Back-End chassis provide the following signals, to be measured with high impedance devices (such as scopes or digital voltmeters):

- Full Bandwidth signal, output: $-10V \dots +10V$;
- Low Pass filter: signal with low/modulator ripple, output: $-10V \dots +10V$, bandwidth: 4.2kHz;
- d/dt: the sensitivity is -10 [Volt/sec/Volt], output: $-5V \dots +5V$.

Analog outputs are also available thru connector P4, at the rear of the Back-End chassis.

FUNCTIONAL TESTS

Observe that the LED 'Range A' is lit.
Make a note of the value displayed on voltmeter or oscilloscope.

Set the calibration switch to 'Calibration Pos.'
Observe the LEDs 'CALON' and 'Range A' are lit.
Observe that the reading increases by $+8$ Volts.

Return the switch to 'Calibration OFF'. Observe the LED 'CALON' goes off.

Set the Calibration switch to 'Calibration Neg.'
Observe that the LEDs 'CALON' and 'Range A' are lit.
Observe that the reading changes by -8 Volts.

Note: if the reading exceeds either $+10.5$ or -10.5 Volts, the sensor may saturate and require demagnetizing and zero adjustment.

Return to Calibration OFF. (The range should still be set to 'Range A')

Set the Test switch to 'Test ON':
Observe that two LEDs 'Test' and 'Range B' are lit.
Observe that the 'zero' reading does not change more than ± 0.5 Volt.

When this sequence of functional tests is completed successfully, it means that the basic functioning of the PCT is correct.

ZERO ADJUSTMENT

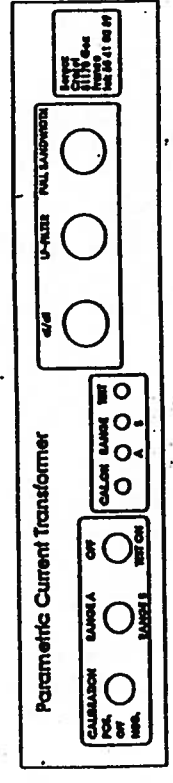
The zero output can be adjusted for a given position of the toroid sensor in respect to its magnetic environment:

- Set the range switch on 'Range A'
- Set the Calibration switch on 'Calibration OFF'
- Set the Test switch 'OFF'
- Remove the M3 screw located on the side of the front-end electronics box
- Insert a screw driver in the screw hole and rotate the 20-turn potentiometer to adjust the reading to zero.

Note that the zero adjustment displaces the working point of the parametric magnetic amplifier and may cause a temporary drift. The instrument should be allowed to stabilize up to a few hours depending on the amplitude of the adjustment.

CONTROL AND STATUS INDICATORS

The back-end electronics box front panel allows complete control of every function of the PCT:



Switches

- 'Calibration' has three positions:
 - OFF: normal operating position
 - POS.: sends a positive calibration current equal to 80% of full scale range A
 - NEG.: sends a negative calibration current equal to 80% of full scale range A
- 'Range' has two positions:
 - Range A: full scale as shown on front page of this manual
 - Range B: full scale as shown on front page of this manual
- 'Test' has two positions:
 - OFF: normal operating position
 - TEST ON: performs an overall loop gain test

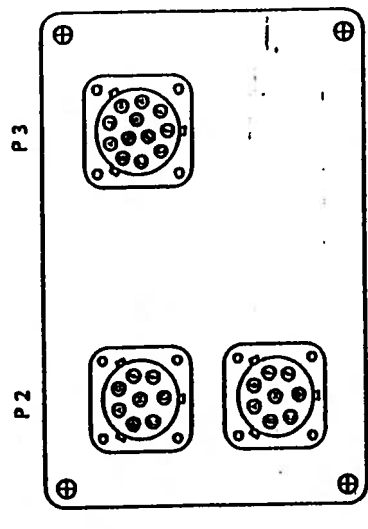
Important: Switch settings are always overridden by control levels applied thru P6 (External Control and Monitoring Connector)

LEDs

- 'CALON': when lit, means calibration positive or negative is applied
- 'RANGE A': when lit, means the instrument is on range A
- 'RANGE B': when lit, means the instrument is on range B
- 'TEST': when lit, means the instrument is performing loop gain tests.

CONNECTORS AND CABLES

Connectors P1, P2 and P3 are located on the front-end electronics box cover



P1 Connector - MODULATOR CABLE

Sensor side Function	Pair	Wire	Cable Belden 8777 3 shielded twisted pairs	Cable connector Bumby Metabloc Baran 8-pin male UTC 612-8PN	Std pin #	CERN pin #	Electronics box side Bumby Metabloc Baran 8-pin female UTC 012-8S	Std pin #	CERN pin #
Static screen		cable shield			A	or	A	or	8
Outer screen		blue			B	or	B	or	2
Free		black			C	or	C	or	3
Sense +		dummy			D	or	D	or	4
Sense -		red			E	or	E	or	5
Excitation 1		red			F	or	F	or	6
Excitation 2		green			G	or	G	or	7
		white			H	or	H	or	1
		black							

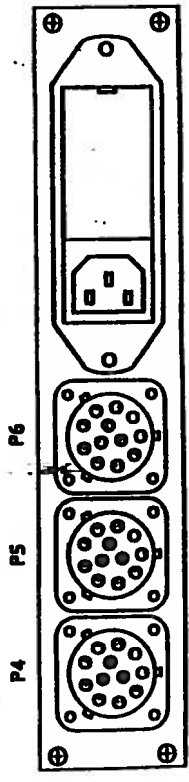
P2 Connector - INTEGRATOR CABLE

Sensor side Function	Cable Belden 8777 3 shielded twisted pairs	Cable connector Burndy Metalok Bantam 8-pin male UTG 612-8PN	Electronics box side Burndy Metalok Bantam 8-pin female UTG 012-8S
Static screen	Pair	Wire	
N/C	cable shield		
Sense +	green	white	
Sense -	red	black	
Feedback +	blue	red	
Feedback -	blue	black	
Calibration +			
Calibration -			

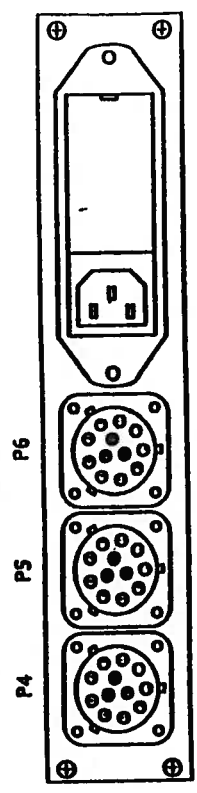
P3 & P5 Connectors - PCT INTERCONNECT CABLE

Front-end electronics box side Chassis connector: Burndy Metalok Bantam 12-pin female UTG 014-12 S	Cable Belden 8777 3 shielded twisted pairs	Cable connector (female) Burndy Metalok Bantam 12-pin male UTG 614-4PN	Back-end chassis side
Function	Pair	Wire	Function
Ground			Ground
Vib output +	N/C		Ground
Vib output -			N/C
+15 V	red	black	Vib output +
-15 V	green	red	Vib output -
Mux'd control	green	white	+15 V
Mux'd status	blue	black	-15 V
Free			Mux'd status
			Free
			Free

Connectors P4, P5 and P6 are located at the back of the back-end chassis:



Connectors P4, P5 and P6 are located on the front-end electronics box cover



P4 Connector - ANALOG OUTPUT CONNECTOR

Function	Cable connector Burndy Metalok Bantam 12-pin female UTG 014-12 S	Mating connector is: Burndy Metalok Bantam 12-pin male UTG 614-12 PN
Vib + differential output	Std pin #	CERN pin #
Vib - differential output	K	1
drift signal	L	2
drift signal ground	M	3
LP-filtered signal	H	4
LP-filtered signal ground	J	5
Full bandwidth signal	A	6
Full bandwidth signal ground	B	7
N/C	C	8
Ground	D	9
Cable screen	E	10
	F	11
	G	12

IMPORTANT: Use only shielded twisted pairs

P6 Connector - EXTERNAL CONTROL AND MONITORING CONNECTOR

Function	Cable connector Burndy Metalok Bantam 12-pin female UTG 014-12 S	Mating connector is: Burndy Metalok Bantam 12-pin male UTG 614-12 PN
Status : Calibration	Std pin #	CERN pin #
Control : Test	K	1
Status : Test	L	2
Control : Range 8	M	3
Status : Range 8 ok	H	4
Control : Range 8	J	5
Status : Range 8 ok	A	6
Control : Range 8	B	7
Status : Range 8 ok	C	8
Control : Range 8	D	9
Status : Range 8 ok	E	10
Control : Range 8	F	11
Status : Range 8 ok	G	12

When signal in is TTL high : PCT sets to Test

When signal in is TTL high : PCT sets to Test

When signal in is TTL high : PCT sets to Test

When signal in is TTL high : PCT sets to Test

When signal in is TTL high : PCT sets to Test

When signal in is TTL high : PCT sets to Test

When signal in is TTL high : PCT sets to Test

When signal in is TTL high : PCT sets to Test

When signal in is TTL high : PCT sets to Test

OPERATING CONSIDERATIONS

Saturating the toroidal sensor

You may accidentally saturate the magnetic material of the sensor's toroidal cores and its ultra high permeability inner shield. This can be caused by:

- Exposing the sensor to magnetic fields higher than those which would provoke a full scale reading.
- Passing a current exceeding the set range by more than 5%.
- Setting the instrument on calibration (positive or negative) when the zero reading is offset by more than 0.5 Volt.

When the sensor has been saturated, the instrument will drift for several days unless you demagnetize it.

Demagnetizing the toroidal sensor

Demagnetizing occurs automatically each time you apply the mains to the instrument.

To demagnetize the sensor:

- Pull the mains plug off the back-end chassis.
- Connect an oscilloscope to the middle BNC labeled 'LP-Filter'.
Set it on 1V/div., 20ms/div.
- Reapply the mains power and observe the demagnetization process on the oscilloscope. It will last ca. 30 seconds depending on sensor's size.

If for whatever reason the power is interrupted during this sequence, the mains power should be reapplied again only after 15 minutes. The zero may need to be readjusted.

Undesired currents

The PCT reads the sum of all currents passing thru the toroidal sensor. Any current flowing thru the vacuum pipe, bake-out sleeve or thermal shield will be 'seen' by the PCT. Make sure the vacuum pipe has an isolating gap to avoid undesired currents and assure that all currents injected into a bake-out sleeve are returned to the same side of the toroidal sensor.

Shorting the toroidal sensor

The sensor must not be shorted. Any loop around the toroid with a loop impedance <1k Ω will cause instability and severe reading errors.

This shorting can occur in unexpected ways, thru additional shields, mechanical holders or braces, thermal shields, etc.

Make sure any conductive part surrounding the toroid does not close a loop thru its center.

Stray fields

The PCT is sensitive to magnetic fields. Its toroidal sensor must be protected against stray fields. To this end, it includes an integrated magnetic shield made out of ultra-high permeability material. A magnetic field of 0.1 A/cm will typically provoke a reading of ca. 2 mA depending on sensor's dimensions. Optionally, some sensors receive an additional composite magnetic shield which provides a shielding factor in excess of 50. It is however the user's responsibility to reduce to an acceptable level the magnetic field to which the sensor is exposed.

Ionizing radiations

The front-end electronics box and the back-end chassis must be protected against exposure to ionizing radiations. Doses as small as 10³ Gy can damage the MOS components used. When the front-end electronics box has to be installed in a locale where significant levels of radiation are present, it is advisable to hide the electronics box in a pigeon hole either in a wall or in the floor.

The toroid sensor contains materials which can be activated by high energy particles, particularly, the sensor contains up to 0.1 kg of Cobalt.

Other materials used in the construction of the sensor can be damaged at high doses.

Radiation sensitive materials used in the "Standard" sensor

The "Standard" sensor is the default sensor unless the "Rad Hard Sensor" option was ordered (see front page of this manual for eventual "Rad Hard Sensor" option)

		Radiation resistance ¹
Wiring insulation	Polyvinylchloride	2×10^5 Gy
	Fiber glass	$> 10^6$ Gy
Stress absorbant	with rubber adhesive	$> 10^6$ Gy
	Silicon rubber tape	5×10^5 Gy
Cables	Silicon rubber	2×10^5 Gy
	Polyvinylchloride	2×10^5 Gy
Insulation Shield Sleeve	Aluminium Polyester	$> 10^6$ Gy
	Polyvinylchloride	2×10^5 Gy
Connector insulation,	Polyphenylene sulfide Ryton-R-4	$> 5 \times 10^6$ Gy

Radiation sensitive materials used in the "Rad Hard" sensor

The "Rad Hard" sensor is only used when this option is ordered (see front page of this manual for eventual "Rad Hard Sensor" option)

Wiring insulation	Polyether-ether-ketone PEEK	6×10^7 Gy
Stress absorbant	Fiber glass	$> 10^6$ Gy
	with rubber adhesive	$> 10^6$ Gy
Cables	Polyurethane foam	5×10^5 Gy
	Polyether-ether-ketone PEEK	5×10^5 Gy
Insulation Static shield Sheath insulation	Copper	6×10^7 Gy
	Polyurethane rubber	no limit
Connector	Polyphenylene sulfide Ryton-R-4	6×10^6 Gy
		$> 5 \times 10^6$ Gy

The above radiation resistance values are only indicative. They do not imply any guarantee of whatever nature from the manufacturer. The manufacturer specifically declines any responsibility for any damage, direct or consequential, caused by ionizing radiations.

¹ Compilation of Radiation Damage Test Data, H. Schottlander et al., CERN 75-04, 75-08, 82-10 and 88-12.

Switch settings and external commands

The instrument can be controlled either from the front panel switches or by TTL levels fed to the instrument's logic thru Connector P3.

In all cases, an external command overrides a switch setting.

The front panel LEDs always display the actual mode of operation of the front-end electronics. The LEDs remain off when the front-end electronics box is not connected to the back-end chassis.

The valid commands are:

"Test ON" when "Calibration Off"
 "Calibration Pos." or "Calibration Neg." when "Range A" and "Test Off"

Some commands combinations are incompatible and would disturb the instrument (ie. sending a full scale calibration current when the instrument is set on range B, etc.). The following combinations are forced, respectively inhibited:

When "Test ON":

Range B is forced to allow maximum sensitivity.
Calibration is inhibited.

if "Calibration Neg.", the range is forced to "Range A" to protect against saturation.

When "Calibration Neg." and "Range B":

Range A is forced
Calibration is set

When "Calibration Pos." and "Range B":

Range A is forced
Calibration is inhibited

This logic prevents incompatible commands.

Julien BERGOZ
01170 Crozet, France
Tel. (33) 50.41.00.89
Fax (33) 50.41.01.99

LIMITED WARRANTY

BERGOZ warrants this instrument to be free from defects in material and workmanship under normal use for a period of three years from the date of shipment. This warranty is exclusive and in lieu of all other warranties expressed or implied. BERGOZ shall not be liable for any consequential, incidental or special damages. Warranty does not cover damages due to ionizing radiations. Faulty instruments must be returned to BERGOZ freight prepaid.

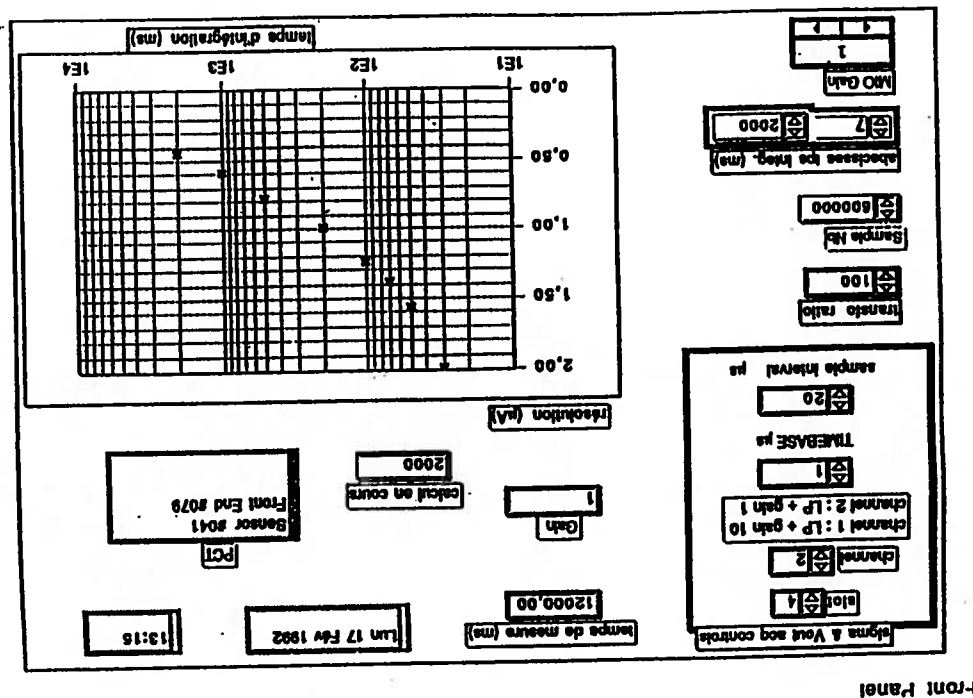
Note that the front-end electronics require specially developed instrumentation for their adjustment. This box does not contain any field serviceable part.

The front-end electronics box is sealed to protect it from any intervention of a non-trained person. Breaking this red seal placed around the front-end electronics box voids the warranty.

ACKNOWLEDGEMENTS

This instrument was designed by Dipl. Ing. Klaus B. Unser of CERN's LEP instrumentation group.

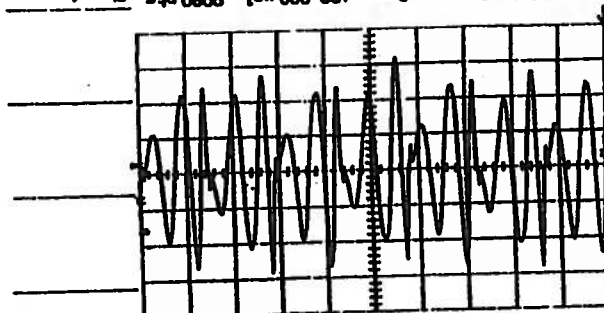
Crozet, December 1991



16:00:8
28-JUL-8

ADDITION

Main Menu



Output Full Bandwidth

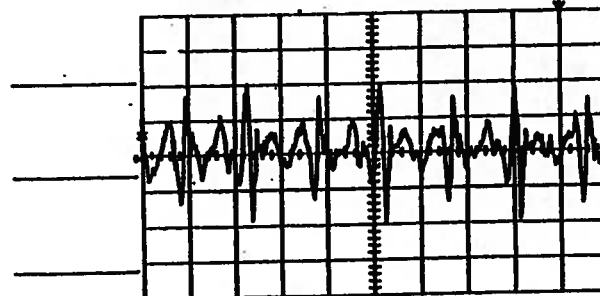
Range B **Ripple**

SENSOR #041

17-05-02
8:28:31

आइए

Model Menu



Output Full Bandwidth

Range A **Ripple**

SENSOR #041

SENSOR #041

Front End #079

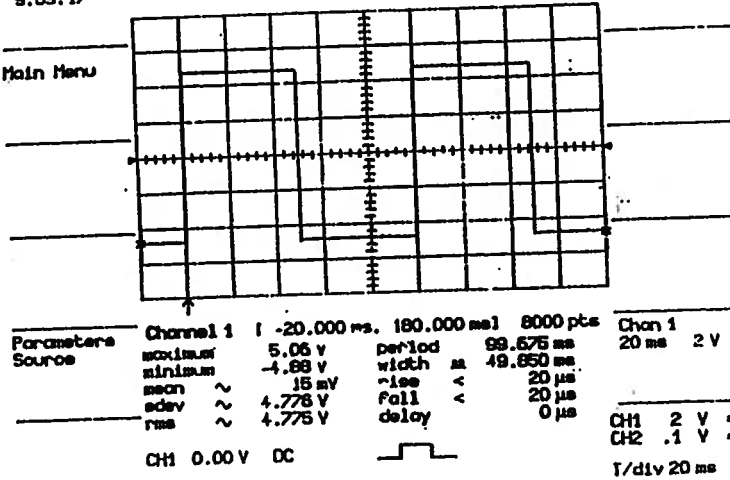
Range A Full scale square wave

Output Full Bandwidth

17-Feb-92
9:59:17

LeCroy

Main Menu



SENSOR #041

Front End #079

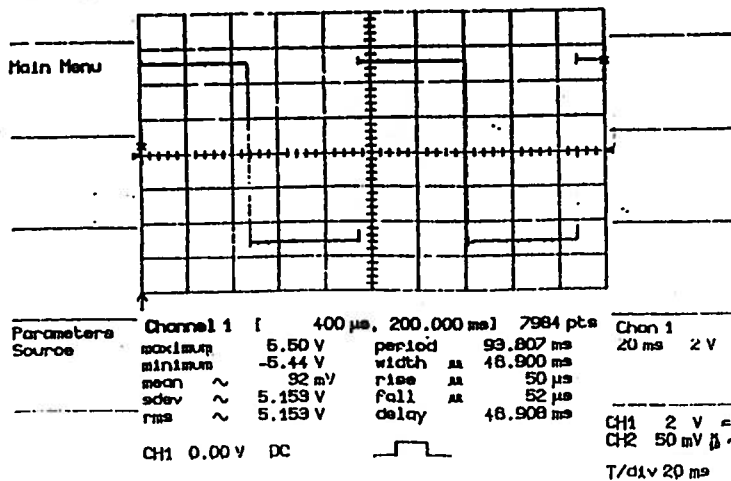
Range B Full scale square wave

Output Full Bandwidth

8-Mar-92
8:19:48

LeCroy

Main Menu



HALL A
CURRENT MONITOR

**Operations
Manual**

Summary

Summary

Description of the System

- Parametric Current Transformer
- Resonant Cavities
- Associated Electronics
- Cables
- Temperature Stabilization system

Block Diagram

Control System Software Description

- Block Diagram
- Hardware
- Epics
- Medm Screens
- Calibration Software

BCM Calibration

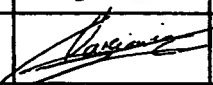
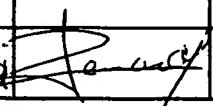
- Preliminary Set-up
- Time to Perform the Calibration
- Procedure for Hall A Operators
- Troubleshooting
- HOW TO REBOOT THE IOC
- Contacts
- How does it Work?
- Note on the High Threshold

List of Medm Screens

- Hall A Current Monitor Screen
- Hall A Current Monitor Small Screen
- Beam Current Monitor Calibration Screen
- High Threshold
- Multimeters Configuration Screen

Annexes

- Calibration Program

	Name	Signature	Date
Author	Geraud Laveissiere		11/25/92
Checked by			
Approved by	J. C. Denard		11/25/97

Description of the System

Parametric Current Transformer

General Description:

Basically, The Direct Current Transformer is composed of two identical toroidal cores driven in opposite ways by an external source. In the absence of any current, the sum of the output signals from the sense windings around each core is zero. A dc beam current through the cores produces a flux imbalance, thus an output signal at even harmonics of the excitation frequency.

The Parametric Current Transformer is usually called "UNSER" from the name of its designer.

Performances & Limitations:

As you directly measure the charge going through the cores, the Unser can be used as an absolute monitor and this is very valuable for a non-destructive current monitor.

However, due to the different magnetic properties of the two toroidal cores, an offset is measured in the absence of any current. In order to make a measurement, one need to subtract this value to the measurement. Now the problem is that the uncertainty on the offset is about 200nA; this is due to temperature, magnetism, microphonics effects, etc. This means that the accuracy will be 1% at 20μA or 1% at 200μA; it is in fact the biggest limitation of the system.

Brief purpose:

The mission of the Unser monitor is to provide an accurate absolute measurement (for example by taking a measurement at high current) in order to calibrate the two resonant cavities located around it.

Calibration:

Instead of measuring the current, you prefer to measure the voltage around a set of precision resistors, well thermostabilized. By this way you need to use a reference to calibrate the system.

To achieve this, a wire has been inserted through the Unser, in which you inject an accurate current produced by a high-precision current source, Keithley 224-2243

Electromagnetic Shieldings:

As the Unser is very sensitive to magnetic fields, it has been surrounded by four shieldings: a Cu electrical shielding, a mumetal magnetic shielding and two Fe magnetic shieldings.

For more information, refer to annexe 1.

Resonant Cavities

General description:

Two resonant cavities are positionned in the beam line behind and in front of the Unser monitor (when you look from the target); they are respectively called "Upstream" and "Downstream" cavities.

These cavities are cylindrical wave guides made out of 304 stainless steel. Their dimensions are 15.48cm in diameter and 15.24cm in length. The resonant frequency is adjusted to the 1497MHz frequency of the Cebaf beam by a stub tuner mounted on a micrometer that can be moved in and out of the cavity. Inside the body, a magnetic field probe, which is a loop antenna, is located on the circumference, where the electrical field is minimum and the magnetic field maximum. The probe is coupled to one of the resonant modes of the cavity and used to couple the beam signal out of the cavity.

Theory of operation:

The electron beam passing through the cavity will excite the resonant transverse electromagnetic mode TM₀₁₀ at 1497MHz and the probe loop will provide an output signal proportionnal to the current. This signal represents in fact the power induced by the beam inside the cavity, and is proportionnal to the Q factor of the cavity. Q is the ratio between the total

energy W stored in the cavity and the total dissipated power P_d , weighted by the resonant frequency ω :

$$Q = \frac{\omega \cdot W}{P_d}$$

or if we assume that the dissipation of the energy occurs only on the walls of the cavity:

$$Q \approx \frac{\text{volume}}{\text{area}}$$

$Q=3000$ for these cavities.

Temperature dependance:

From the equations one see that Q is sensitive to the resonant frequency of the cavity and consequently to the dimensions of the cavity. Since the change of the dimensions is related to the variation of the temperature, to perform an accurate measurement one must be sure that the temperature is constant.

Position dependance:

The power coupled from the beam to the cavity at radius ρ also depends for the TM010 mode (this mode is relatively insensitive to the position of the beam) to the variations of the electric field along the z axis $E_{z\rho}$:

$$P = \text{const} \cdot I_{\text{beam}} \cdot E_{z\rho}$$

thus the variation of E_z from the change in the beam position represents the position dependance of the current measurement.

The effect of the beam displacement is about 0.06% /cm.

Performances & Limitations:

Unlike the Unser the cavities are not absolute measurements, they are only relative because you collect a signal which is proportionnal to the beam current. so they need to be calibrated using an absolute measurement like the Unser monitor.

However, as they are very stable and linear, once they have been well calibrated, that is you have transfered to them the accuracy of the Unser, you can use them for the regular measurement on all the dynamic range.

Associated Electronics

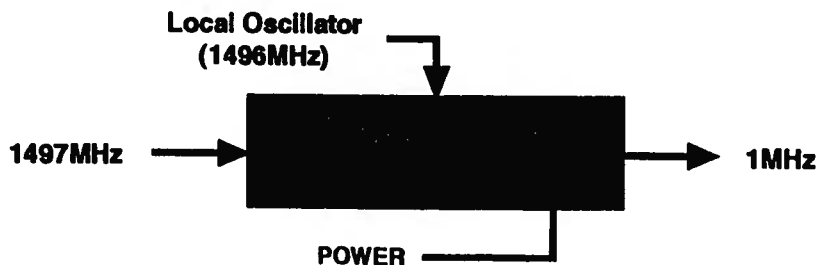
Unser electronics:

Below the grey box in the hall is located the Front-End Electronic of the Parametric Current Transformer. As it is located 1m below the beam line, this black box is shielded with lead bricks. This box needs to be removed to adjust the offset of the Unser (see Annexe 1).

The second part, called Back-End Electronic, is located in the counting house in rack #CH01B06. From there all the controls can be operated manually.

Down-Converters:

The purpose of this system is to downconvert a high-frequency signal (1497MHz RF) in an intermediate frequency signal (1MHz IF for example). The input RF signal is mixed with a local oscillator signal to get the output IF signal:



$$v_{RF}(t) = \cos(\omega_{RF}t + \theta_{RF}) \quad \text{input RF signal}$$

$$v_{LO}(t) = \cos(\omega_{LO}t + \theta_{LO}) \quad \text{local oscillator}$$

$$v_{IF}(t) = v_{RF}(t) \times v_{LO}(t) \quad \text{output IF signal}$$

$$= \frac{1}{2} \{ \cos[(\omega_{RF} + \omega_{LO}) + (\theta_{RF} + \theta_{LO})] + \cos[(\omega_{RF} - \omega_{LO}) + (\theta_{RF} - \theta_{LO})] \}$$

The residual 3GHz component is eliminated by a filter (see Schematic 6).

By skin effect, a 1.5GHz signal gets quickly attenuated in the cables (the effect is proportional to the square of the frequency). The down-converter allows you to carry the signal over long distances without any problem.

Nevertheless, some aspects limit the capabilities of the system: first, the gain (typically +4dBm) is a function of time and temperature, and second, the linearity of the system is limited, at low level by the external noise (60Hz), at high level by the saturation of the down-converter itself.

The two down-converters are installed in Hall A behind the shield wall, in a Euro 3U crate.

RMS to DC converters:

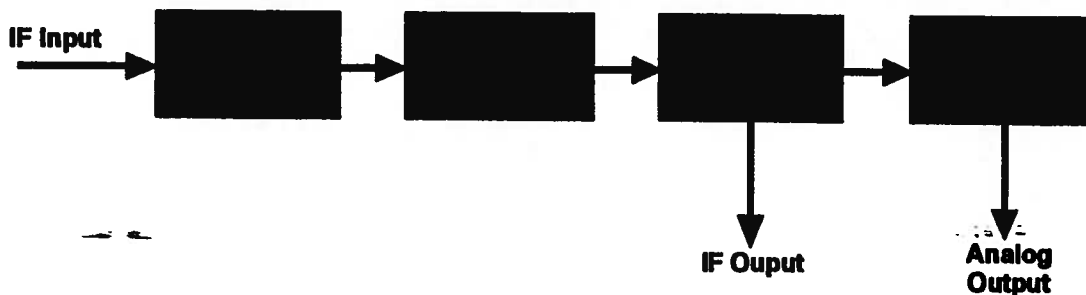
The goal of this system is to filter, amplify and convert the output of the down-converter into a continuous signal proportional to the rms value of the input. The system is also designed to produce some alarms in case of any problem, for example a loss of power, or an invalid connexion.

The system is composed of several steps:

- a first amplifier which makes sure the linearity is good over all the dynamic range
- a dual amplifier band pass filter which eliminates any noise carried along with the input signal, and whose center frequency is adjusted during the calibration to the output of the down-converter
- an amplifier with adjustable gain
- the RMS to DC chip itself that makes the conversion, with an adjustable offset.

(see schematic 8)

In our case, the system has been slightly modified in order to get two different output signals: the first one is picked after the second amplifier and is still an IF signal, the second one is the regular dc output. For regular measurements only the first signal is used and the RMS to DC is only used as a filter. For certain experiments, like parity violation, the dc output is necessary to get a signal which is directly proportional to the integrated charge.



Cables

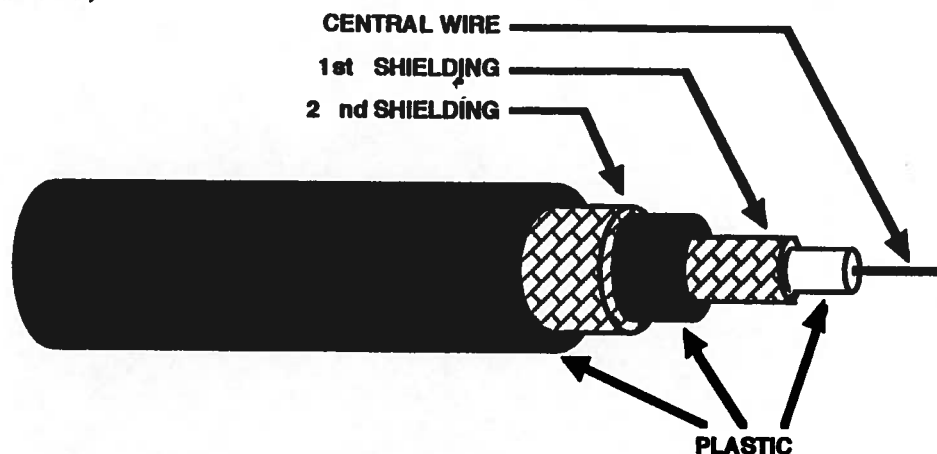
Heliac 7/8":

The cables CH01-5212 and CH01-5213 are made with the Andrew Heliac 7/8" part# LDF5-50A. These cables are about 50 feet long and run from the BCM box to the downconverters chassis behind the shielding wall.

Refer to the catalog for more information on this cable and the connectors.

Triaxial:

The cables CH01-5209, CH01-5214-1, CH01-5214-2 are made with the Belden triaxial cable part #9222 (refer to catalog for more information). The connectors for these cables are the ITT Pomona 2 lug male cable plug triaxial connectors part #885-5056 (refer to catalog for more information).



As this type of cable is not common at Jefferson Lab, above is a drawing of the section of the cable to fix the ideas.

Temperature stabilization system

Thermocouples:

To regulate the temperature inside the box, 6 T-type thermocouples have been installed at different places (see diagram #x):

- inside the aluminum cover
- on the Unser monitor
- in contact with the copper shield which surrounds the Unser
- on the Upstream cavity
- on the downstream cavity
- outside the box (monitors the Hall temperature)

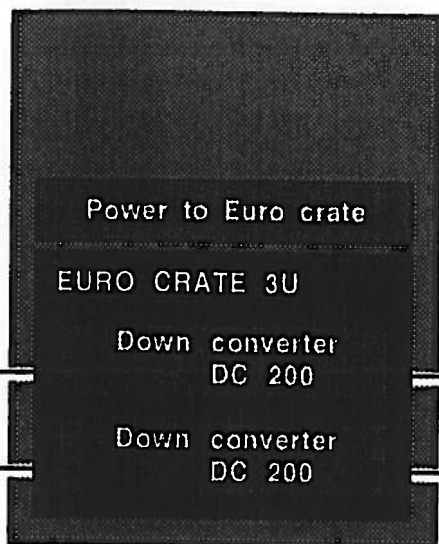
these six thermocouples are connected to a temperature control chassis.

Temperature control chassis:

As the most temperature-sensitive device is the Unser monitor, the first thermocouple is used for the regulation and is connected to the temperature controller which adjusts the current in the heat tapes.

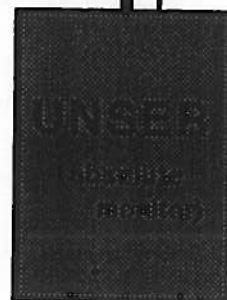
The other thermocouples are connected to a readback display.

Rack in Hall A



helix 1/2'
CH01-E213

sma cable
CH01-5211



sma cable
CH01-5210

PCT
front
end

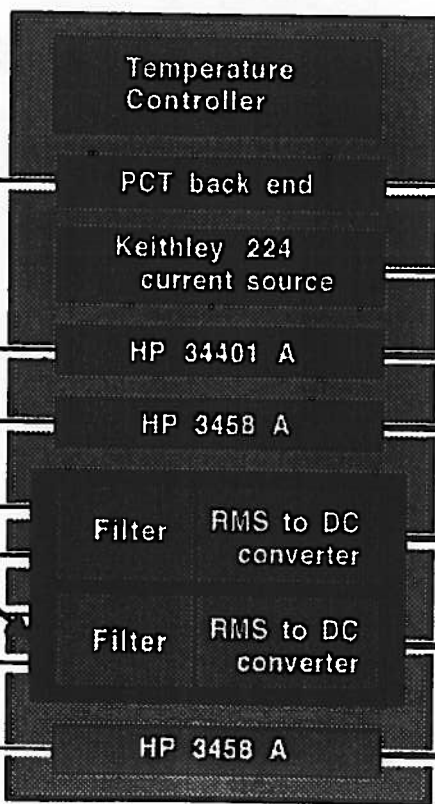
triax cable
CH01-5214-1

triax cable
CH01-5214-2

helix 1/2'
CH01-5212

junction
box

Rack In Hall A counting house



BEAM

CH01-5203

triax cable
CH01-5203

junction
box

chassis

junction
box

Digital output

Digital
output

Analog
output

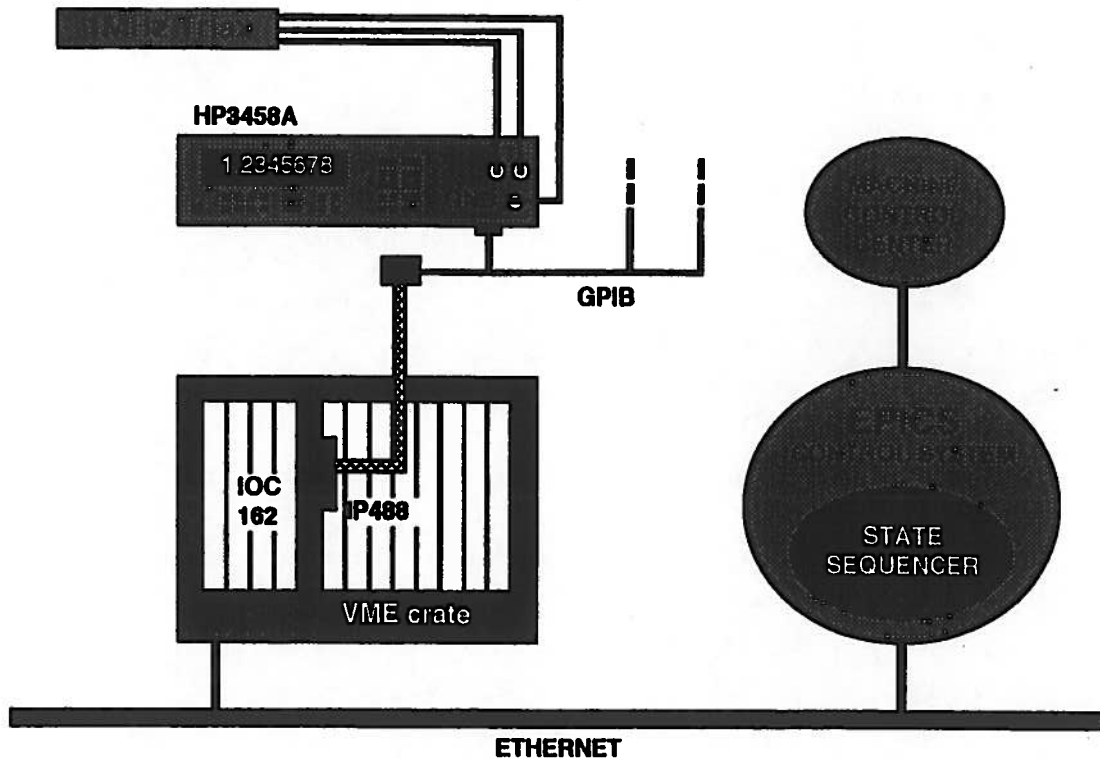
Analog
output

Digital
output

Hall A Current
Monitor
Block Diagram

Control System Software Description

Block Diagram



Hardware

GPIB:

The three multimeters are controlled via the Hewlett Packard GPIB system. The bus is used to send commands to the devices and also to receive the measurements.

IP488:

The GPIB cables are connected with a 50pins ribbon to the Industry Pack IEEE488.

MVME162:

The IOC where the IP488 is plugged, is a MVME162 located into a regular VME crate in the counting house.

Epics

General Description:

When the IOC boots, it loads Epics, the appropriate database and the State Sequencer Program into itself; From then users are able to control the three digital multimeters and record informations from them.

The Multimeters are triggered once per second by Epics; they are also configured to make a one second integrated measurement; thus the current is measured all the time. However remember that it doesn't mean the values are recorded all the time by the Data Acquisition System (CODA).

Contact Persons:

All the controls are done in Epics; for more information contact a specialist:
Javier Gomez, Yves Roblin.

Medm Screens

General Control:

The main screen displays the intensity read by the monitor numerically, and on a strip chart so that people can visualize the evolution of the current in the machine. To rescale the stripchart, type the new low and high limits of the desired scales, then close and open the screen again, the stripchart will appear with the new ranges.

This screen also displays the elapsed time since the last calibration of the monitor. When this time becomes greater than 1 day, a red blinking message appears which says "CALIBRATION REQUIRED".

From this screen you can access to the multimeters configuration and calibration screens. also, by clicking on the Instructions button, you can call a file containing all the basics of the current monitor operations.

General Control - Small Screen:

This screen is just a small version of the Main Control screen and doesn't include a stripchart.

Multimeters Configuration:

This screen allows you to change the configuration of the three digital Voltmeters. The left part is connected to the Upstream bcm, the middle part to the Parametric Current Transformer and the right part to the Downstream bcm.

For the two HP3458A, the default parameters for normal operation (called when the IOC is rebooted) are:

- FUNC ACV
- SETACV SYNC
- Range AUTO
- Autozero ON
- Resolution 0.01%
- Triggering Mode AUTO
- AC Band Low 999900
- AC Band High 1000100
- Sampling Mode AUTO
- Level Filter OFF
- Buffering OFF
- Number of Events 1
- Reading ON

All these parameters are setup during the initialization phase and should not be modified during normal operation. Some of them are highlighted in red on the screen

For the HP34401A the parameters for normal operation are:

- FUNC DCV
- Range AUTO
- Autozero ON
- Input Impedance OFF

Calibration Operations:

A special screen allows the users to setup for calibration operation. Simples control of the three devices are also provided:

- for the Unser monitor, the "adjust coefficient" control allows you to slightly change the Unser coefficient around the nominal value of 250; the "subtract offset correction" control allows you to apply an extra offset to the reading of the Unser; this cannot be used during calibration operation in order to not interfere with the process.

- for the two cavities, the "on/off" button suspends the reading process of the multimeter (useful to change the configuration); Although the Coefficients will be updated at the end of any calibration, you are allowed to change them as needed.

- for all three devices, the voltage read by the multimeter as well as the corresponding current are displayed.

An other box controls the calibration process: to begin it, click on the "start" button; "status" goes from "stopped" to "in process"; each time another step is reached, the "Step number" is implemented by 1; to stop it, click on the "stop" button, "status" goes back to "stopped", the "time since last calibration" is reset to zero, and the "calibration required" message stops blinking.

As for the main control screen, you can call the other related screens and a brief operation tutorial.

High Threshold:

A very small screen that allows the operator to change the calibration high threshold.

Calibration Software

The calibration is performed by a State Sequencer Program loaded into epics when the IOC is booted.

See next chapter for more details, and the annexe for a program listing.

BCM Calibration

The Hall A Beam Current Monitor Calibration is the main operation the operators have to perform.

Preliminary Set-up:

- On HAC, open a new terminal
- type "HAC_hp" or "HAC_xt" on this terminal; this will open the Hall A main Control screen
- Select Beamline → Current monitor to open the Hall A Current Monitor screen
- From the Hall A Current Monitor screen, pull the "Calibration" screen.
- From the Calibration screen, pull the "Threshold" screen.

Time to perform the calibration:

The calibration needs to be done approximately **twice every day** to ensure a good accuracy of the measurements. It cannot be done during a run.

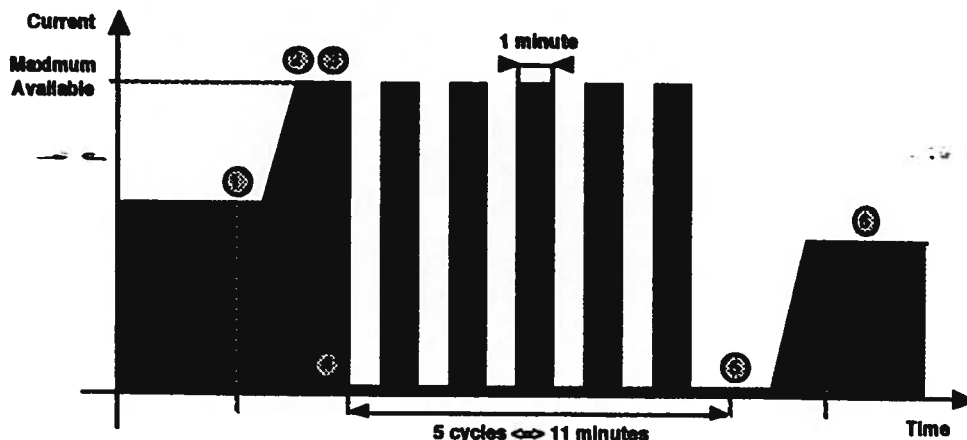
12 hours after the last calibration, a blinking alarm signal appears on both screens (Hall A Current Monitor & Calibration).

If another Hall is running at the same time and has priority, the calibration must be negotiated with these people or be performed during an access of this Hall, because the procedure requires to stop the beam.

BCM CALIBRATION FOR HALL A & C CAN BE DONE AT THE SAME TIME.

Procedure for Hall A operators:

- ① Ask MCC to setup the maximum current they can deliver in Hall A; accuracy of the calibration will be $\approx 1\% \times 20 / I_{\text{max}}(\mu\text{A})$.
- ② Adjust the Threshold to 80% of the DVM output voltages (3V for 80 μA).



- ③ When ready, ask MCC to start the Hall A BCM calibration procedure

- ④ On the calibration screen, push "START" button: the red message "Stopped" should turn into "In Process"; the number of steps should get initialized to 0.
- ⑤ After 5 steps (corresponds to about 10min) push "STOP" button: the red message "In Process" should turn into "Stopped"; the coefficients and their associated standard deviations should be updated.
- ⑥ Paste the Calibration screen into the Log book.

Troubleshooting:

During a normal operation, the coefficients of calibration are updated after each step; The two important rules to check if the calibration is valid are:

- The coefficients should be fairly identical (± 2).
- They should be between 20 and 26.

Remember also that whatever happens, during normal operation (except when accelerator is in pulsed mode) the reading of the current on the Current Monitor Screen should be updated every second.

PROBLEM	SOLUTION
One coefficient is completely out of range (for example 0.000 or 43457.987)	Check if the readings (uA) of the two cavities are still updated every second. If one value is stocked and doesn't update, REBOOT the ioc.
The two coefficients are not identical but still in the right range (for example 21.00 and 29.67)	Check the readings in V of the two cavities; if they are identical then stop the calibration process and try it again. If they are still different, REBOOT the IOC.
One coefficient is bad, but not that bad (for example 17.56)	There is a software problem: please call an expert. If he doesn't answer, you can either REBOOT the IOC and start another calibration, or replace the bad coefficients by the old ones.
One coefficient doesn't even indicate a number (for example ??0.00)	If you don't want to redo a calibration, replace the nasty number by the corresponding previous coefficient.
One of the two cavity readings doesn't update itself anymore	REBOOT the IOC.
The medm screen freezes and becomes blank (no control)	REBOOT the IOC.
The number of steps is not incremented during the calibration.	REBOOT the IOC and retry the calibration from the beginning. If it fails again, call an expert.
Nothing works	Call an expert

HOW TO REBOOT THE IOC:

THE IOC IS LOCATED IN A VME CRATE IN RACK CH01B06 (THE FIRST RACK IN THE MIDDLE ROOM OF THE HALL A COUNTING HOUSE). IN ORDER TO REBOOT IT, YOU HAVE THE CHOICE BETWEEN:

- PUSH THE RESET/ABORT BUTTONS ON THE IOC ITSELF (AT THE SAME TIME)
- PUSH THE RESET BUTTON ON THE VME CRATE (CAREFUL, THE BUTTON HARDLY COMES BACK)
- TURN THE CRATE OFF, LEAVE IT OFF 1MINUTE, AND TURN IT BACK ON.

IF ALL THESE PROCEDURES FAILED, CALL AN EXPERT.

Contacts:

In case of any problem, especially if the troubleshooting procedures failed, don't hesitate to call an expert:

- Javier Gomez x7498
- Branko Vlahovic x6331 pager 680 6331
- Yves Roblin x7481
- Geraud Laveissiere x5032 pager 680 0079

How does it work?

If you really want to know all the secrets of the calibration, here they are.

The calibration is performed by a "state sequencer program" (C based) which runs inside the IOC. The program works in several steps:

- first the program loads the previous values of the calibration coefficients and the last calibration time from files, and inserts them into Epics.
- then it waits until the operator pushes the START button; during this time, the program updates the elapsed time since the last calibration, and initiates a blinking message in case this time gets over 12 hours.
- once the START button is pushed, the program initializes all the variables and then goes in "beam" state.
- In this state, the program checks every 0.2 seconds whether the current values have been updated, or whether the STOP button has been pushed.

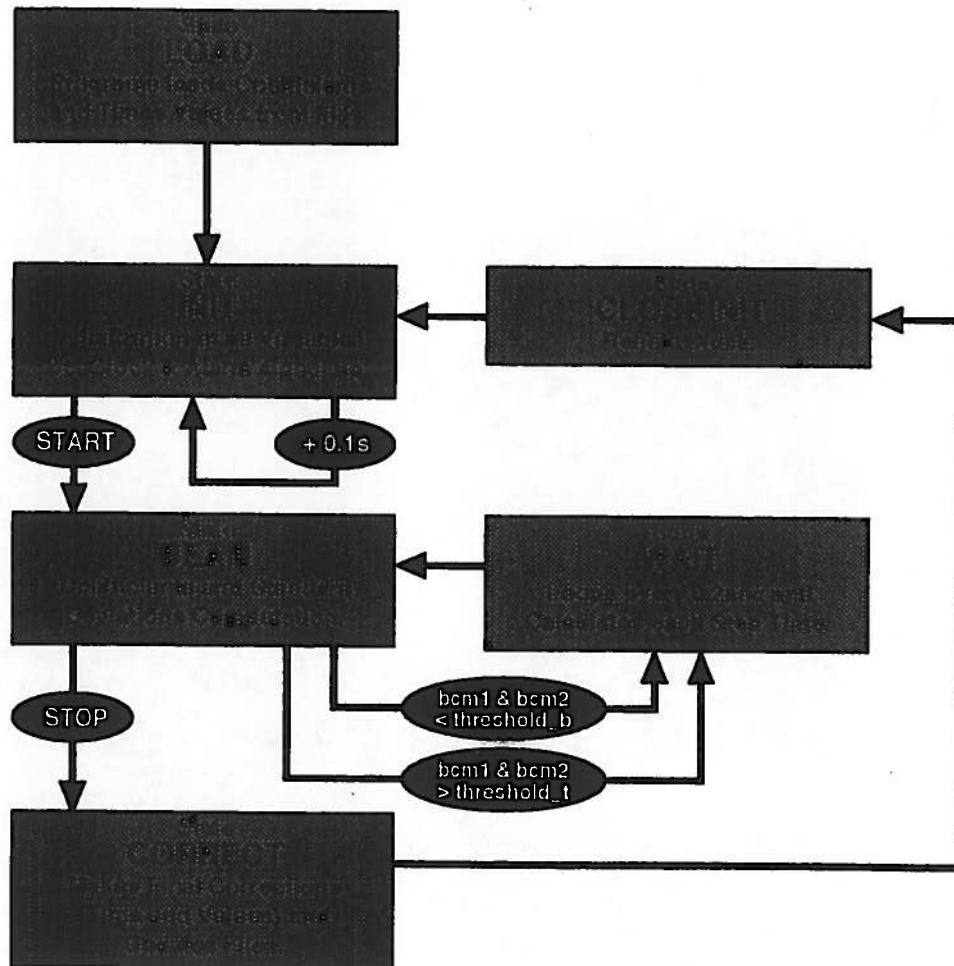
In the first case, the program checks the presence of the beam by comparing the values from the bcms to a low and a high threshold: if the values are below the low threshold, that means there is no beam and the program has to make an Unser offset measurement; if the values are above the high threshold, that means the beam is here and the program has to make the 3 current measurement (Unser + 2 cavities). Each time the beam status changes, the program calculates the coefficients associated with the previous step, and updates the Epics database.

In case the STOP button is pressed, the program calculates the last step and the final standard deviations.

- Finally the program performs a correction on the coefficients: it recalculates the steps average but this time forgets about the bad steps (a step is considered bad if it is more than 5% above or below the nominal value).

- Finally the program resets the intern clocks and goes back to step #2.

State Sequencer Program Logical Diagram



Note on the High Threshold:

I am using the thresholds method because I want to avoid being corrupted by the intermediate values when the beam is shut off or when it comes back in the bcm. It is most important that the High threshold is set to a proper value, because it tells the sequencer whether the beam is present or not.

List of Medm Screens

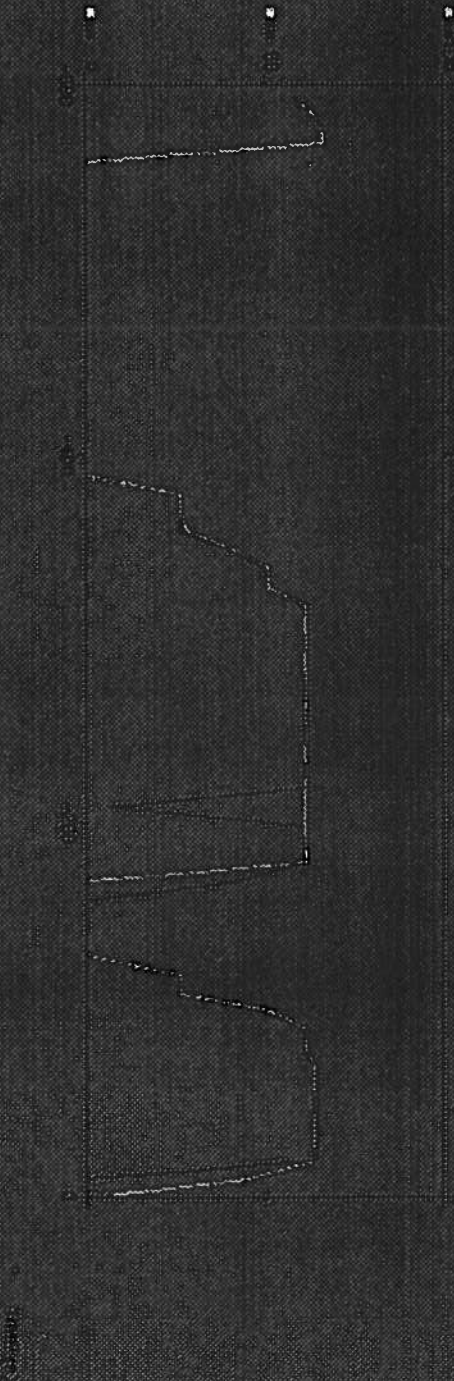
Screen Name	File Location
Hall A Current Monitor	/hac_epics/disp/bcm/Hac_bcm.adl
Hall A Current Monitor	/hac_epics/disp/bcm/Hac_bcm_small.adl
Beam Current Monitor Calibration	/hac_epics/disp/bcm/Calibration.adl
High Threshold	/hac_epics/disp/bcm/threshold.adl
Multimeters Configuration	/hac_epics/disp/bcm/Multimeters.adl

The second screen is a small version of the general Hall A Current Monitor screen and doesn't contain any stripchart.

Hall A Current Monitor

CURRENT (mA)

HALL A CURRENT



UNITS

mA

0 100

Current

mA

0 100

Output

mA

0 100

Calibration

1

Time since last calibration

0 days 0 hrs 00 min

CALIBRATION

HE COMPLETED

CURRENT (mA)

0.033

Hall A Beam Current Monitor

OPERATION

0.849

These values are calibration

Range 10 nA 68 nA

Calibration Required



Beam Current Monitor Calibration

Upper Monitor

0.00535 V

-0.837 uA

Correct Beam Current

29

0.000

Upstream Cavity

1.932427 V

46.527 uA

23.352

0.29

Downstream Cavity

1.863053 V

45.644 uA

24.488

0.30

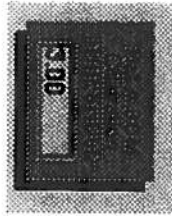


Calibration

start stop

Instructions

Related Screens



HP3458A AC Voltmeter Configuration

Upstream BCM

The image displays a variety of electronic components, primarily integrated circuits (ICs) and passive components like capacitors and resistors. The components are arranged in a grid-like fashion, with labels and markings visible on their surfaces.

ICs and Modules:

- Top Left:** A large IC with a label "RESET" and "DCV". Below it, another IC is labeled "ACV".
- Top Center:** A module labeled "AUTO" with a display showing "1", "10", "100", and "1000".
- Top Right:** A module labeled "OFF" with a display showing "ON" and "ONCE".
- Middle Left:** A module labeled "AUTO" with a display showing "EXT", "LVL", "LINE", and "HOLD".
- Middle Center:** A module labeled "AUTO" with a display showing "EXT", "LVL", "LINE", and "HOLD".
- Middle Right:** A module labeled "AUTO" with a display showing "EXT", "LVL", "LINE", and "HOLD".
- Bottom Left:** A module labeled "AUTO" with a display showing "EXT", "LVL", "LINE", and "HOLD".
- Bottom Center:** A module labeled "AUTO" with a display showing "EXT", "LVL", "LINE", and "HOLD".
- Bottom Right:** A module labeled "AUTO" with a display showing "EXT", "LVL", "LINE", and "HOLD".

Passive Components:

- Capacitors:** Several electrolytic capacitors are visible, with values like "1000", "100", "10", and "1" marked on them.
- Resistors:** Various resistors are scattered throughout, with values like "100", "10", "1", and "0.0000" marked on them.

The components are arranged in a grid-like fashion, with labels and markings visible on their surfaces. The labels include "RESET", "DCV", "ACV", "AUTO", "EXT", "LVL", "LINE", "HOLD", "ON", "ONCE", "1", "10", "100", "1000", "0.00", "0.0000", "0", and "1".

HP34401A

DC Voltmeter Configuration

Unge

Figure 1 shows the front panel of the 1000 Series. It features a 'RESET' button at the top right. Below it is a digital display showing '0.0'. To the left of the display is a 'POWER' switch with 'OFF', 'ON', and 'ONCE' positions. Below the display is a 'FUNCTION' selector with 'AUTO', '10', '0.1', '100', '1', and '1000' settings. At the bottom are two 'HOLD' buttons and a 'MODE' button.

HP3458A AC Voltmeter Configuration

Downstream BCM

The image displays a variety of electronic calculators, including scientific and basic models. The calculators are arranged in a grid, showing different brands and features. Some have multiple display windows, while others have single displays. The buttons are typically black with white or yellow markings. The overall appearance is that of a product catalog or a collection of vintage electronic devices.

ANNEXES

program calibrate

```
/* This state sequencer program is designed to perform the Hall A Beam Current Monitor Calibration */
/* To compile it, type "snc calib.st" (this converts it into a .c file) then you have to type "make" to have it compile and ready to be loaded */
/* The program is loaded into the IOC Hallasc5 memory and executed by Epics */

/* All files are located in: */
/* "/hac_epics/ioc/hallasc5/epics/seq/" . */

/* Written in 1997 by: */
/* Yves Roblin */
/* Geraud Laveissiere */

/* ***** */
/* INCLUDE LIBRARIES ***** */
/* ***** */

%% #include <stdio.h>
%% #include <stdlib.h>

/* ***** */
/* INTEGER DECLARATIONS ***** */
/* ***** */

/* IMPORTANT ***** */
/* u corresponds to the Unser monitor */
/* 1 corresponds to the Upstream BCM */
/* 2 corresponds to the Downstream BCM */
/* o corresponds to the Unser monitor offset */
/* ***** */

/* Step Variables */
/* indexes of current values */
int i; /* Unser,bcm1,bcm2 */
int io; /* Unser offset */
/* numbers of stored values */
int j; /* Unser,bcm1,bcm2 */
int jo; /* Unser offset */

int k;

/* Step time limits = 20sec/1minute */
int timelimit;

/* position in step 0=offset 1=top */
int position;
/* number of steps */
int step;
/* authorize computation of coefficients 0=yes 1=no */
int compute;
/* authorize update of date 0=no 1=yes */
int update;

Control Button */
0 = Calibration OFF */
/* 1 = Calibration ON */
int done;
```

```

/* Unser,BCM1,BCM2 readings */
float bcm1;
float bcm2;
float unser;

/* Final coefficients and standard deviations */
float A1;
float A2;
float sdev1;
float sdev2;

/* Low threshold that defines offset computation */
/* It is set at the beginning of the program. */
float threshold_b;

/* high threshold that defines Unser and bcms computation */
/* It is defined by user through another screen. */
float threshold_t;

/* ***** */
/* FILES POINTERS DECLARATIONS ***** */
/* ***** */

%% FILE *fpcoef;
%% FILE *fpdate;
%% FILE *fpthres;

***** */
ASSIGN VARIABLES TO CHANNELS ***** */
***** */

assign done to "hac_bcm_done";
assign uncalibrated to "hac_bcm_uncalibrated";
assign blicky to "hac_bcm_blicky";
assign bcm1 to "hac_bcm_dvm1_read";
assign bcm2 to "hac_bcm_dvm2_read";
assign unser to "hac_unser_current";
assign A1 to "hac_bcm_A1";
assign A2 to "hac_bcm_A2";
assign sdev1 to "hac_bcm_A1sdev";
assign sdev2 to "hac_bcm_A2sdev";
assign step to "hac_bcm_count";
assign swmin to "hac_bcm_swmin";
assign swhr to "hac_bcm_swhr";
assign swday to "hac_bcm_swday";
assign threshold_t to "hac_bcm_threshold_t";

/* ***** */
/* CHANNEL ACCESS MONITORING ***** */
/* ***** */

monitor unser;
monitor done;
monitor bcm1;
monitor bcm2;
monitor threshold_t;

/* ***** */
/* BEGINNING OF PROGRAM ***** */

```

```

swsec=0;
swmin=0;
swhr=0;
swday=0;
pvPut (swmin);
pvPut (swhr);
pvPut (swday);
}
} state init
}

```

```

/* ***** */
/* INITIALIZATION STATE: */
/* The sequencer checks continuously the value of the button "done" */
/* and updates the clock every second. Every 9/10 second, it also */
/* updates the blinking signal by switching the value of "blicky" */
/* from 0 to 1. */
/* Every minute, the program writes the present time into the file */
/* "calib_date_file". */
/* In case the START button is pressed (done=1) the program starts */
/* the calibration process. In order: */
/* - initialize A1,A2,sdev1,sdev2 */
/* - initialize values1,values2,correction1,correction2 */
/* - initialize correct1,correct2 */
/* - initialize step */
/* - initialize position */
/* - initialize compute */
/* then it goes to beam state. */
/* ***** */

```

```

state init {

```

```

    when (done == 1) {
        printf("starting calibration\n");
        A1=0.0;
        A2=0.0;
        sdev1=0.0;
        sdev2=0.0;
        for (k = 1;k <= 25;k++) values1[k]=0.0;
        for (k = 1;k <= 25;k++) values2[k]=0.0;
        for (k = 1;k <= 25;k++) correction1[k]=0.0;
        for (k = 1;k <= 25;k++) correction2[k]=0.0;
        for (k = 1;k <= 25;k++) steptime[k]=0;
        correct1=0;
        correct2=0;
        step=0;
        printf("step #%d\n",step);
        pvPut (step);
        position=2;
        compute=1;
    } state beam

```

```

when (delay(0.1)) {
    swsec++;

    /* start blinking process */
    if (uncalibrated==1) {
        swblicky++;
        if (swblicky == 9) {
            swblicky=0;

```

```

/* in order to understand this part of the program. */
/* ***** */

state beam {

/ ***** */
/* USER PUSHES STOP BUTTON: */
/* First the program checks whether the number of steps is */
/* more than zero, because you don't want to perform anything */
/* in case of a phantom calibration. */
/* Then it computes the last step (update of values1 & 2) and */
/* updates the coefficients and their standard deviations on */
/* the screen. */
/* Finally, it resets the variables uncalibrated and blicky, */
/* and jumps into the last state: correct. */
/* ***** */

    when (done == 0) {
        printf("calibration stopped\n");

        /* compute A1 and A2 coefficients */
        if (step > 0) {

            steptime[step + 1]=timelimit;

            values1[step] = modif1;
            values2[step] = modif2;
            printf("modif1=%f modif2=%f",modif1,modif2);
            printf(" bcm1=%f bcm2=%f\n",A1 * bcm1,A2 * bcm2);

            printf("computing coefficients\n");
            /* updating coefficients */
            A1=A1/step;
            A2=A2/step;
            pvPut (A1);
            pvPut (A2);

            for (k = 1;k <= step;k++) {
                sdev1 = sdev1 + (values1[k]-A1)*(values1[k]-A1);
                sdev2 = sdev2 + (values2[k]-A2)*(values2[k]-A2);
            }
            sdev1 = sqrt( sdev1/step );
            sdev2 = sqrt( sdev2/step );
            pvPut (sdev1);
            pvPut (sdev2);

            printf("A1=%f +/-%f  A2=%f +/-%f\n",A1,sdev1,A2,sdev2);

            uncalibrated=0;
            pvPut (uncalibrated);
            blicky=0;
            pvPut (blicky);

        }
        printf("end of process\n");
    } state correct

/ ***** */
/* BEAM IS PRESENT (Faraday Cup 2 is OUT): */
/* First the program determines whether: */

```

```

/*      is made only by 1 measurement.                                */
/* For each bcm the actual calculation is made by:                    */
/* - first substracting the Unser Offset (calculated during          */
/*   last step) from the Unser measurement                            */
/* - dividing the average of the Unser measurements during           */
/*   the step by the average of the BCM measurements during          */
/*   the step.                                                         */
/* *****                                                             */

/* check if values changed */
    if (unser != topu[i]) {
/* don't compute if more than one minute has ellapsed */
        if (timelimit > 280) {
            if (i == j) i=2;
            else i++;
            compute=1;
        }

        else {
            i++;
            j=i;
        }
/* calculate new values */
        s1=s1 - top1[i];
        s2=s2 - top2[i];
        su=su - topu[i];
        if (i != 1) {
            top1[i]=bcm1;
            top2[i]=bcm2;
            topu[i]=unser;
        }
        s1=s1 + top1[i];
        s2=s2 + top2[i];
        su=su + topu[i];
    }

/* compute new coefficient */
    if ( (compute == 0) && (i != 1) ) {
        A1=A1 - modif1;
        A2=A2 - modif2;
        modif1=( (su/(j - 1))-(so/(jo - 1)) ) / (s1/(j - 1));
        modif2=( (su/(j - 1))-(so/(jo - 1)) ) / (s2/(j - 1));
        A1=A1 + modif1;
        A2=A2 + modif2;
    }

    if (timelimit == 280) printf("stop computing\n");
    } state wait

/* *****                                                             */
/* BEAM IS NOT DETECTED (Faraday cup 2 is IN):                        */
/* First the program determines whether:                              */
/* - we are at the beginning of the calibration (position=2)         */
/*   In this case, the program just resets all the variables        */
/*   i,j,s1,s2,su,top1,top2,topu,modif1,modif2,timelimit and        */
/*   sets the variable position to "0". Number of step also         */
/*   gets increased by 1.                                             */
/* - or we are at the beginning of a new step (position=1)           */
/*   Here the program performa the same initialization as in         */
/*   the first case, and also computes the previous step: it        */
/*   updates the variables values1 & 2 and A1 & 2.                  */
/* - or beam was already present (position=0)                         */
/* Then the program starts or goes on computing the step.            */

```

```

        compute=1;
    }
    else {
        io++;
        jo=io;
    }
/* calculate new values */
    so=so - offset[io];
    if (io != 1) {
        offset[io]=unser;
    }
    so=so + offset[io];
}

/* compute new coefficient */
if ( (compute == 0) && (io != 1) ) {
    A1=A1 - modif1;
    A2=A2 - modif2;
    modif1=( (su/(j - 1))-(so/(jo - 1)) ) / (s1/(j - 1));
    modif2=( (su/(j - 1))-(so/(jo - 1)) ) / (s2/(j - 1));
    A1=A1 + modif1;
    A2=A2 + modif2;
}
if (timelimit == 280) printf("stop computing\n");
    } state wait
}

/ ***** */
CORRECTION STATE: */
At the end of the Calibration, the program reviews all the */
intermediate coefficients, checks them, and if necessary redoes */
/* the final computation for A1,A2,sdev1,sdev2, taking off the bad */
/* steps. */
/* ***** */

state correct {
    when () {
        if (step > 0) {

/* ***** */
/* Here it checks whether the intermediate steps belongs to */
/* some confidence interval: */
/* [19 27] for the Upstream BCM */
/* [20 28] for the Downstream BCM */
/* For each bcm, in case one coefficient is bad, the */
/* corresponding value of the variable correction1 (or 2) is */
/* set to 1 (instead of 0); also the value of correct1 (or 2) */
/* is not 0 anymore. */
/* also the program checks whether the duration of each step */
/* was more than 20sec and make the appropriate correction. */
/* Finally, we know which coefficients are bad by the variable */
/* correction1 (or 2). */
/* ***** */

        for (k = 1;k <= step;k++) {
            if (values1[k] < 19) {
                correction1[k] = 1;
                correct1++;
            }

```

```

    }

/* ***** */
/* If some correction has to be made, the program recomputes sdev1 and/or sdev2. */
/* ***** */

    if (correct1 != 0) {
        sdev1=0;
        for (k = 1;k <= step;k++) {
            if (correction1[k] != 1) sdev1=sdev1+(values1[k]-A1)*(values1[k]-A1);
        }
        sdev1=sqrt( sdev1/(step-correct1) );
        pvPut(sdev1);
    }

    if (correct2 != 0) {
        sdev2=0;
        for (k = 1;k <= step;k++) {
            if (correction2[k] != 1) sdev2=sdev2+(values2[k]-A2)*(values2[k]-A2);
        }
        sdev2=sqrt( sdev2/(step-correct2) );
        pvPut(sdev2);
    }

/* ***** */
/* Finally the program updates the file that contains A1,A2, sdev1,sdev2, and the file that contains threshold_t. */
/* ***** */

    fpcoef = fopen("/hac_epics/ioc/hallasc5/epics/seq/calib_coef_file"
"%f %f %f %f",A1,sdev1,A2,sdev2);
    fprintf(fpcoef,"%f %f %f %f",A1,sdev1,A2,sdev2);
    fclose(fpcoef);
    fpthres = fopen("/hac_epics/ioc/hallasc5/epics/seq/calib_threshold"
"%f",threshold_t);
    fprintf(fpthres,"%f",threshold_t);
    fclose(fpthres);

    }
    } state clockinit
}

}

/* ***** */
/* END OF PROGRAM ***** */
/* ***** */

```

These documents will replace the corresponding pages when the 10kHz system will be operationnal, year 1998.

For more details, see John Musson.

energy W stored in the cavity and the total dissipated power P_d , weighted by the resonant frequency ω :

$$Q = \frac{\omega \cdot W}{P_d}$$

or if we assume that the dissipation of the energy occurs only on the walls of the cavity:

$$Q \approx \frac{\text{volume}}{\text{area}}$$

$Q \approx 3000$ for these cavities.

Temperature dependance:

From the equations one sees that Q is sensitive to the resonant frequency of the cavity and consequently to the dimensions of the cavity. Since the change of the dimensions is related to the variation of the temperature, to perform an accurate measurement one must be sure that the temperature is constant.

Position dependance:

The power coupled from the beam to the cavity at radius ρ also depends for the TM010 mode (this mode is relatively insensitive to the position of the beam) to the variations of the electric field along the z axis E_z :

$$P = \text{const.} \cdot I_{\text{beam}} \cdot E_z$$

thus the variation of E_z from the change in the beam position represents the position dependance of the current measurement.

The effect of the beam displacement is about 0.06% /cm.

Performances & Limitations:

Unlike the Unser the cavities are not absolute measurements, they are only relative because you collect a signal which is proportionnal to the beam current. so they need to be calibrated using an absolute measurement like the Unser monitor.

However, as they are very stable and linear, once they have been well calibrated, that is you have transferred to them the accuracy of the Unser, you can use them for the regular measurement on all the dynamic range.

Associated Electronics

Unser electronics:

Below the grey box in the hall is located the Front-End Electronic of the Parametric Current Transformer. As it is located 1m below the beam line, this black box is shielded with lead bricks. This box needs to be removed to adjust the offset of the Unser (see Annexe 1).

The second part, called Back-End Electronic, is located in the counting house in rack #CH01B06. From there all the controls can be operated manually.

Down-Converters:

The purpose of this system is to downconvert a high-frequency signal (1497MHz RF) in an intermediate frequency signal (10kHz IF for example). The input RF signal is mixed with a local oscillator signal to get the output IF signal:

